ENGREQ

ISEP - Mestrado em Engenharia Informática

Bio Cantinas - Software Development Report Sprint 2

2026

Bio Cantinas

SOFTWARE DEVELOPMENT REPORT – SPRINT 2

DIOGO RODRIGUES 1221959

1 Introduction 4
1.1 Purpose 5
1.2 Scope sprint 2 5
1.3 References 6
2 Work Summary 7
2.1 Overview of the activities performed in Sprint 2 7
2.2 Key outcomes and deliverables produced 8
3 Conceptual Domain Model 9
3.1 Menu and Meal Management 9
3.1.1 Main entities: 9
3.1.2 Key relationships: 10
3.2 Customer, Reservation, and Feedback Management 10
3.2.1 Main entities: 10
3.2.2 Key relationships: 10
3.3 Planning and Order Management 11
3.3.1 Main entities: 11
3.3.2 Key relationships: 11
3.4 Supply Chain and Inventory Management 11
3.4.1 Main entities: 11
3.4.2 Key relationships: 11
3.5 Supplier Recruitment and Onboarding 12
3.5.1 Main entities: 12
3.5.2 Key relationships: 12
3.6 Monitoring, Statistics, and Data Persistence 12
4 Component Architecture 13
4.1 Component diagram 13
4.2 Responsibilities and interactions between components 15
5 User Story Mapping 17
5.1 Per User Story 17
5.3 MVP 18
5.4 MVI1 19
5.5 MVI2 20
6 Backlog 21
PAGEREF _Toc216734468 \h 7 .... Non-Functional Requirements 23

8 Application Visual Evidences 25
9 Version Control Evidence 37
10 .... Conclusion 39

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1 Introduction

The BioCantinas project aims to support the sustainable management of school and nursing home canteens by promoting the use of organic and locally produced food in the Cinfães Municipality.

This document describes the software development for the BioCantinas web application, which will support the main processes of menu planning, supplier and stock management, and meal reservation and payment for end users.

1.1 Purpose

This document reports the work carried out during Sprint 2 (from 15 December 2025 to 11 January 2026) of the BioCantinas information system project for the municipality of Cinfães.

The purpose of Sprint 2 was to extend and consolidate the functional MVP delivered in Sprint 1 by implementing the priority user stories defined for this iteration, with particular focus on sanitary control measures (supplier quarantine and parish sanitary shutdown) and on KPI visualization and analysis at different operational levels (canteen, dining hall, and network). Sprint 2 also incorporated stakeholder feedback gathered during the Sprint 1 presentation, and addressed any unfinished functional and non-functional requirements from the previous iteration, ensuring that requirements are satisfied as an integrated whole through the delivered product.

In addition, this report provides evidence and documentation of the sprint outcomes, including the domain model, component diagram, story mapping with:

- Minimum Viable Product (MVP);
- Minimum Viable Increment 1 and 2 (MVI1/MVI2);
- Backlog separation;

Along with a prioritized backlog with assigned responsibilities and justification, application screenshots, non-functional requirements, and version control/repository evidence. The product backlog used to plan and track the sprint is maintained in the project board shown in the provided screenshot.

1.2 Scope sprint 2

Sprint 2 focused on extending and consolidating the solution delivered in Sprint 1, by implementing the Sprint 2 user stories defined in the assignment and incorporating stakeholder feedback.

The core scope for Sprint 2 included:

- Sanitary quarantine and shutdown measures: enabling the canteen manager/network administrator to place a supplier under quarantine (so their products are excluded from meal planning) and to place a “freguesia” (parish) under sanitary shutdown (so products from suppliers located in that parish are excluded while the shutdown is active).

- KPI visualization and filtering: enabling different roles to access KPIs at the appropriate level, including canteen manager and cafeteria manager viewing KPIs for their own unit, and the network administrator viewing KPIs for the entire network with the ability to refine results by canteen, cafeteria, and producer.

- Continuous improvement and completion: addressing feedback from the Sprint 1 presentation and completing any unfinished functional and non-functional requirements from previous work, ensuring that requirements are delivered coherently as part of an integrated product. Some of the improvement additions were aimed at increasing data consistency and decision quality, such as:

    - Supplier application form: validated location selection: enforcing municipality/parish selection (no free-text), making parish selection mandatory during producer register/edit, and ensuring consistent location data.

    - Improved location visibility: showing associated municipalities and parishes, with emphasis on the CIM Tâmega e Sousa region, and loading/displaying zones within a parish (dropdown/list) so users can confirm the correct zone.

    - Menu workflow enhancement: adding the menu status "Waiting for production plan" to better support operational planning.

    - More accurate organic information on bio-product for each meal: improving organic product indicators by displaying organic quantities by weight (e.g., *120g organic out of 500g total*), avoiding misleading interpretations when only¢¢¢¢ minor ingredients are organic.

1.3 References

[1] Municipality of Torres Vedras. Biocanteens. Available at: https://www.cm-tvedras.pt/educacao/saude-e-alimentacao/biocantinas
(Accessed: 15-12-2025).

[2] Polytechnic Institute of Viseu. Good Practices Manual for Sustainable Canteens. Viseu, 2021. Available at: https://repositorio.ipv.pt/bitstreams/download
(Accessed: 15-12-2025).

[3] European Commission. French biocanteens seek to spread organic local food from school canteens across Europe. Available at: https://ec.europa.eu/regional_policy/pt/projects/Romania/french-biocanteens-seek-to-spread-organic-local-food-from-school-canteens-across-europe
(Accessed: 15-12-2025).

2 Work Summary

2.1 Overview of the activities performed in Sprint 2

During Sprint 2, was used the same process used in Sprint 1. The team organized the work using the Product Backlog board (statuses such as "Backlog", "In progress", "In review", "Done" and "Block") and continued the implementation of the BioCantinas system, with continuous integration through GitHub commits and merges.

Building on the Sprint 1 baseline repository and documentation (GitHub repository - https://github.com/Departamento-de-Engenharia-Informatica/ENGREQ_51 , including the setup instructions, routes, and initial credentials), this spring focused on extending the platform according to the Sprint 2 requirements and the feedback received during the Sprint 1 presentation. Functionally, Sprint 2 work concentrated on:

- Sanitary quarantine and shutdown controls: implementing the ability for the canteen manager and/or network administrator to place a supplier under quarantine and to place a parish (freguesia) under sanitary shutdown, ensuring that products from affected suppliers/areas are excluded from meal planning and production while restrictions are active.
- KPI visualization and refinement: extending KPI features so that canteen managers and dining hall managers can view KPIs for their unit, while the network administrator can view KPIs across the network and filter/refine results by canteen, dining hall, and producer.
- Completion and improvement work: addressing any unfinished functional/non-functional items from earlier work and integrating stakeholder feedback to improve the overall coherence of the delivered product.

In addition to the Sprint 2 core scope, the sprint included targeted improvements aimed at data quality and decision accuracy, such as: enforcing validated producer location selection (mandatory parish (freguesia) selection with official lists rather than free-text), displaying municipalities and freguesias with emphasis on the CIM Tâmega e Sousa region, loading zones within a freguesia for confirmation, introducing the menu status “Waiting for production plan”, and improving the interpretation of organic usage by presenting organic products by weight (e.g., *120g organic out of 500g total*) to avoid misleading conclusions when only minor ingredients are organic.

From the technological part, Sprint 2 continued with the same layered web architecture as Sprint 1, using React and Vite on the frontend and Node.js and Express on the backend, exposing REST APIs, with persistence in MySQL supported by Sequelize. Postman was used for endpoint testing, and GitHub remained the central platform for collaboration and traceability.

Overall, Sprint 2 consolidated the MVP and extended the platform with sanitary control mechanisms and advanced KPI capabilities, incorporating stakeholder feedback and maintaining continuous validation with the client with the final objective of a complete functional application.

2.2 Key outcomes and deliverables produced

Sprint 2 delivered an extended MVP with sanitary control features (supplier quarantine and freguesia sanitary shutdown) and improved KPI visualization, enabling KPI access for canteen and cafeteria managers and network-level KPI analysis with filtering (canteen, cafeteria, producer). The sprint also incorporated key improvements such as validated producer location selection (freguesia + zones), the menu status “Waiting for production plan”, and a more accurate organic indicator by showing organic products by weight.

Overall, Sprint 2 progressed the platform towards the final objective of a complete and fully functional application.

3 Conceptual Domain Model

The conceptual domain model illustrated in Figure 1. represents the main entities and relationships that compose the BioCantinas management system. The diagram provides a conceptual view of how meals, reservations, stock, suppliers, recruitment, and operational roles interact throughout the system.

The primary entities include School Customer, Nursing Home Staff, Reservation, Meal, Dish, Ingredient, Menu, Chef, Dietician Team, Canteen, Stock, Stock Manager, Order, Ordered Product, Supplier, Expected Product, Application, Recruitment Team, Batch, Product, and various classification entities (e.g., Type of Meal, Type of Dish, Type of Canteen, Type of Product).

These entities are organized into several functional contexts that structure the overall operation of the BioCantinas system, including meal planning, food preparation, customer reservations, inventory management, and supplier onboarding.

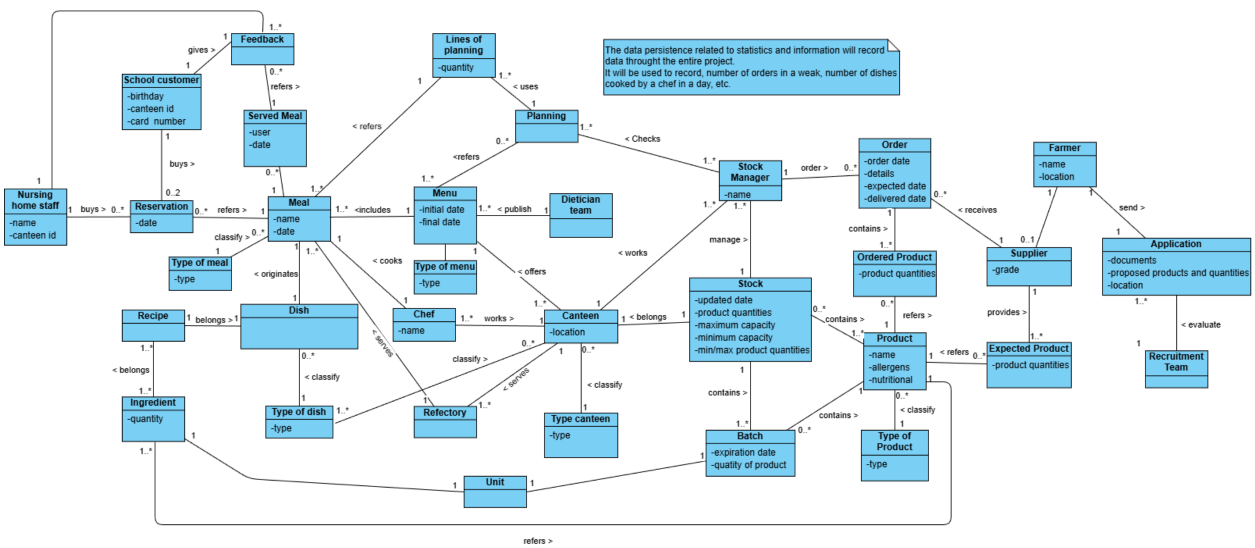


Figure 1 - Conceptual Domain Model

3.1 Menu and Meal Management

This context covers the definition, planning, preparation, and classification of meals and menus in BioCantinas.

3.1.1 Main entities:

- Menu
- Meal
- Dish
- Recipe
- Ingredient
- Chef
- Dietician Team
- Canteen
- Refectory

3.1.2 Key relationships:

- Dietician Teams publish Menus.
- Menus include Meals.
- Dish originate Meals.
- Dishes contain Recipe.
- Recipe contain Ingredients.
- Chefs prepare dishes.
- Canteens offer menus.
- Canteens provide the meals to the refectory.
- Meals are consumed in the refectory.

3.2 Customer, Reservation, and Feedback Management

This context focuses on customer interaction with the system.

3.2.1 Main entities:

- School Customer
- Nursing Home Staff
- Reservation
- Served Meal
- Feedback

3.2.2 Key relationships:

- Customers books a meal.
- Booking a meal originates a Reservation.
- Reservations refer to Meals.
- Served Meals are associated with users.
- Feedback refers to Served Meals.

3.3 Planning and Order Management

This context addresses meal quantity forecasting.

3.3.1 Main entities:

- Planning
- Lines of Planning

3.3.2 Key relationships:

- Planning uses Lines of Planning.
- Planning checks stock availability.

3.4 Supply Chain and Inventory Management

This context covers stock and order control.

3.4.1 Main entities:

- Stock Manager
- Stock
- Product
- Batch
- Order
- Ordered Product
- Supplier
- Expected Products

3.4.2 Key relationships:

- Stock Managers manage Stock.
- Stock contains Batches.
- Batches contains Products.
- Orders contain Ordered Products.
- Supplier gets orders.
- Orders are made using planning, stock and expected products.

3.5 Supplier Recruitment and Onboarding

This context manages supplier evaluation.

3.5.1 Main entities:

- Farmer
- Application
- Expected Product
- Supplier
- Recruitment Team

3.5.2 Key relationships:

- Farmers submit Applications.
- Recruitment Teams evaluate Applications.
- Approved Farmers become Suppliers.

3.6 Monitoring, Statistics, and Data Persistence

The system stores historical and operational data for reporting and analysis.

4 Component Architecture

The system is structured according to a Layered Architecture, which promotes separation of concerns, maintainability, and scalability.

Each layer has a well-defined responsibility and communicates only with its adjacent layers.

The main components of the system are:

- ***Frameworks / Drivers Layer***
  Handles technical concerns such as HTTP routing, database connectivity, and framework configuration.
  It forwards incoming requests to the appropriate adapters and provides infrastructure support to the system.

- ***Adapters Layer***
  Responsible for handling incoming client requests, validating input data, and returning appropriate responses.
  This layer does not contain business logic; instead, it delegates processing to the Service layer.

- ***Application Layer***
  Responsible for accessing the database and performing CRUD operations.
  It orchestrates operations, applies validations, and coordinates interactions between repositories and domain entities.

- ***Domain Model Layer***
  Contains the core business rules of the application.
  Defines the domain entities and data models used throughout the system.

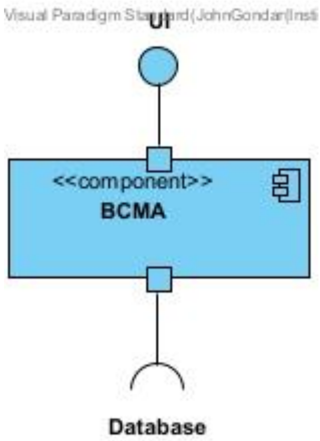
This architecture ensures that changes in one layer have minimal impact on others, making the system easier to extend and maintain.

4.1 Component diagram

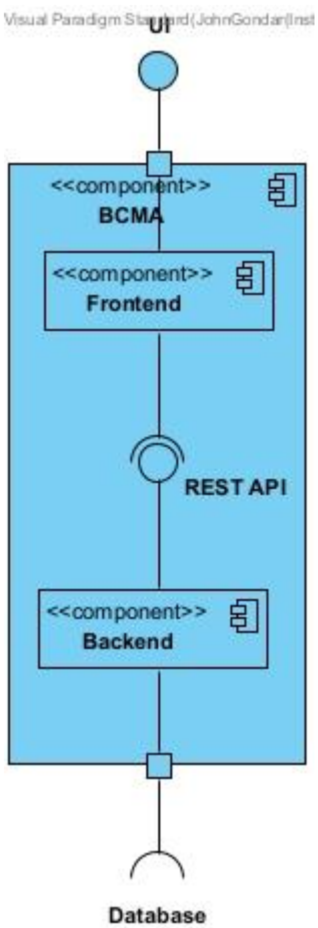
Logical View:

The logical view of the component diagram illustrates the high-level organization of the system components and their relationships, independent of implementation or deployment details. As it's possible to check from the images above, we implemented a layered architecture.

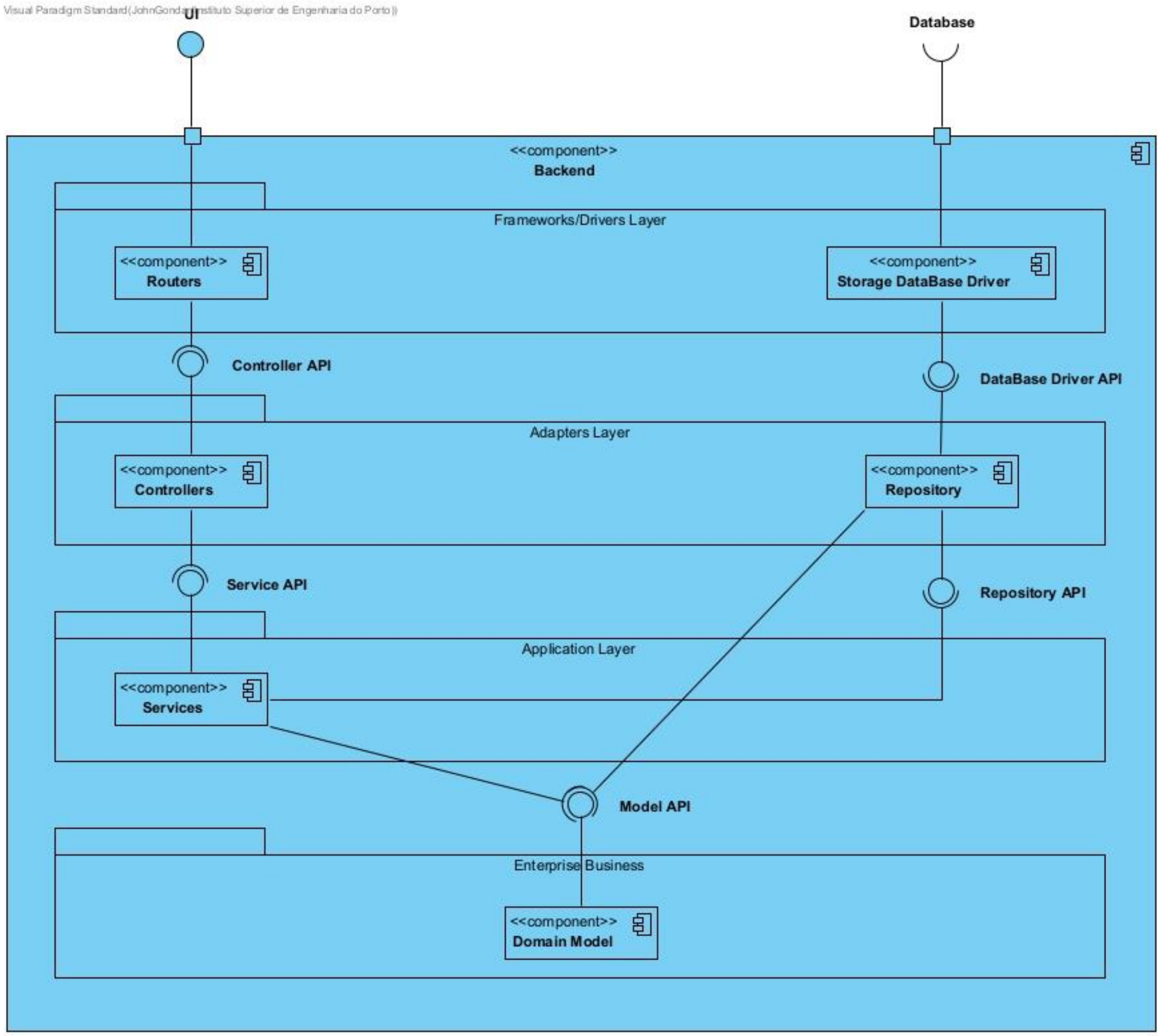
Level 1



Level 2



Level 3



4.2 Responsibilities and interactions between components

- ***Controllers***
  Receive HTTP requests from the client, extract and validate request parameters, and invoke the appropriate service methods.
  After receiving the result from the Service layer, they format and return the response.

- ***Services***
  Implement the business logic of the application.
  Services validate business rules, combine data from multiple repositories when necessary, and decide how operations should be executed.

- ***Repositories***
  Handle all data persistence concerns.
  They interact directly with the database and return domain entities or data objects to the Service layer.

- ***Domain Models***
  Represent the business entities and encapsulate the data structure of the application.
  They are used across the Service and Repository layers.

The interaction flow is strictly top-down:
the Presentation layer communicates with the Service layer, which in turn interacts with the Repository layer.
This clear separation improves testability, readability, and long-term maintainability of the MVP.

5 User Story Mapping

	Access and security	Supplier Management	Recipe Management	Logistics and ordering	Optimization and analysis
Sprint 1					
User story	US1	US2, US4, US6	US3, US5, US11	US10, US12, US9	US7, US8, US13
Sprint 2					
User story		US1, US2		US3	US4, US5, US6

5.1 Per User Story

User Story	Meaning	Responsible	Priority
Sprint 1			
US1	As a user, I want to log in to the application	Diogo Rodrigues	MVP
US2	As a farmer, I want to apply for the position of supplier.	Diogo Rodrigues	MVP
US3	As a nutritionist, I want to publish a meal plan.	Diogo Rodrigues	MVP
US4	As a network manager, I want to approve an application.	Vanessa Pereira	MVP
US5	As a nutritionist, I want to be able to change and approve the nutritional information and allergenic ingredients of a recipe.	Leonardo Costa	MVI2

US6	The system must sort the producers.	Vanessa Pereira	MVI1
US7	The system should calculate the percentage of organic products in a recipe.	João Gondar	MVI2
US8	The system must calculate the percentage of meal waste.	João Fraga	MVP
US9	The system must make suggestions for possible menus.	João Gondar	MVI1
US10	The system must calculate the quantity of products to order and make this forecast available to suppliers.	Diogo Rodrigues	MVP
US11	The system must present nutritional information and allergenic ingredients for the recipes.	Leonardo Costa	MVI1
US12	The system must adjust the necessary quantities of products to order after meal reservations.	Diogo Rodrigues	MVP
US13	The system must generate significant KPIs.	João Fraga	MVI2
Sprint 2			
US1 (User story added based on feedback from Sprint 1)	The supplier application form validates the producer location through parish selection.	Vanessa Pereira	MVP
US2	As a canteen manager / biocanteen network administrator, I want to place a supplier in quarantine so that the products from this supplier are not used in the preparation of meals	Diogo Rodrigues	MVP

	while the quarantine is active.		
US3	As a canteen manager / biocanteen network administrator, I want to place a parish under sanitary lockdown, so that products from suppliers in that parish are not used in meal preparation while the lockdown is active.	João Gondar	MVP
US4	As a canteen manager, I want to be able to view the KPIs of my canteen.	João Fraga	MVI1
US5	As a refectory manager, I want to be able to view the KPIs of my cafeteria.	João Fraga	MVI1
US6	As a biocanteen networkmanager, I want to be able to view the KPIs of the entire network and refine them by canteen, refectory, and supplier.	João Fraga	MVI2

5.3 MVP

MVP stands for Minimum Viable Product, which is the smallest set of features required to create a complete product that delivers core value and can be launched to gather validated learning.

The initial core functionality must be built first to achieve the product's primary goal, which is why we selected US1 (Login), US2 (Supplier application), US3 (Publish meal plans), US4 (Approve applications), and US8 (Calculate meal waste), US10 (Calculate product order quantities and provide forecasts to suppliers), US12 (Adjust product quantities after meal reservations). These User Stories ensure that users can access the system, suppliers can be onboarded, meal plans can be created and managed, applications can be approved by managers, and basic metrics on meal waste can be gathered. Together, they provide a functional that delivers value to end-users while minimizing development effort.

Sprint 1	
US1	Login is fundamental for identifying users and controlling access. Without it, the system cannot differentiate the different users.
US2	This allows the system to onboard suppliers, which are responsible for providing products.
US3	Publishing meal plans is a core function of the system and is required to provide value to end-users.

US4	Enables the network manager to validate and approve farmer applications, allowing them to transition into suppliers responsible for providing products.
US8	Provides basic metrics to monitor efficiency and resource usage, useful for early feedback.
U10	Without this functionality, the system would not support real operational workflows between meal planning and suppliers.
U12	Ensures that product orders remain aligned with actual demand as meal reservations change. It is essential for maintaining accuracy and preventing over-ordering or shortages in the MVP.
Sprint 2	
US1(feedback)	Essential to ensure that all suppliers are correctly registered; prevents errors in the supplier network.
US2	Place a supplier in quarantine. Ensures food safety by preventing products from problematic suppliers from being used in meals.
US3	This functionality also ensures food safety by preventing potentially unsafe products from suppliers in a parish under sanitary lockdown from entering meals.

5.4 MVI1

MVI stands for Minimum Viable Increment, which refers to a subsequent delivery of features that extend the system's capabilities beyond the MVP by adding additional value and improving quality. MVI1 focuses on enhancing usability, decision support, and food safety by introducing functionalities related to supplier organization, menu intelligence, **and** nutritional transparency. For this reason, we selected US6 (Sort producers), US9 (Suggest possible menus), and US11 (Present nutritional and allergenic information).

Together, they strengthen the reliability and usefulness of the system without compromising the core workflows already delivered in the MVP.

Sprint 1	
US6	Enhances user experience by organizing producers for easier access and decision-making.
US9	Adds intelligent recommendations to assist nutritionists and improve meal planning.

US11	Increases transparency and safety for end-users consuming the meals, important but not blocking the MVP.
Sprint 2	
US4	Allows the canteen manager to monitor the performance of their canteen, identify issues early, and make informed operational decisions.
US5	Allows the refectory manager to monitor the performance of their refectory, track efficiency, and support local decision-making.

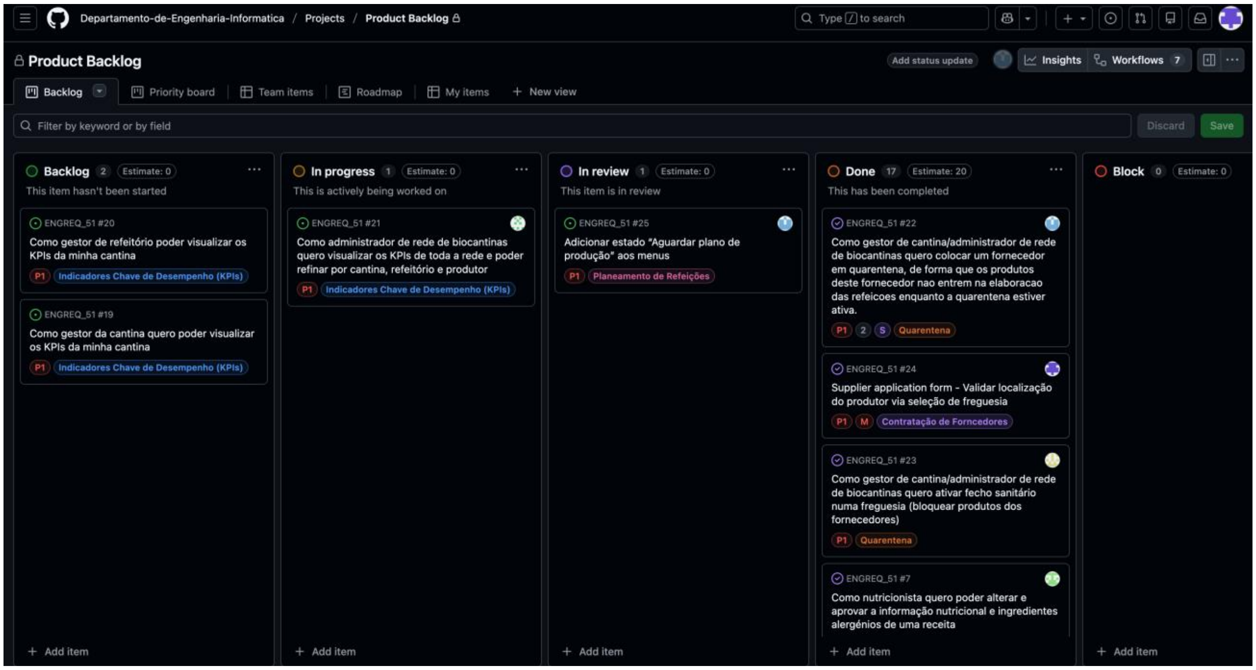
5.5 MVI2

MVI2, the second Minimum Viable Increment, defines the features to be developed after the stabilization of the core product delivered in the MVP and MVI1. This iteration focuses on advanced functionality aimed at optimization, operational efficiency, and strategic decision-making. For this reason, it includes US5 (Manage and approve nutritional and allergenic information), US7 (Calculate percentage of organic products), and US13 (Generate KPIs).

Together, they contribute to a more scalable, data-driven, and fully featured system, while remaining outside the scope of the initial MVP due to their higher complexity and analytical nature.

Sprint 1	
US5	Provides professional control over recipes and ensures accuracy, important for a complete system.
US7	Adds analytics for sustainability and marketing insights, valuable for reporting but not essential initially.
US13	Provides strategic insights and monitoring, valuable for management but not required for initial operation.
Sprint 2	
US6	This functionality gives the network manager a strategic overview of the entire biocanteen network, supporting analysis and optimization, but is not critical for initial operation.

6 Backlog


Departamento-de-Engenharia-Informatica / Projects / Product Backlog
Type / to search
Product Backlog
Add status update
Insights
Workflows 7
Backlog
Priority board
Team items
Roadmap
My items
New view
Filter by keyword or by field
Discard
Save
Backlog 2 Estimate: 0
This item hasn't been started
ENGREQ_51 #20
Como gestor de refeitório poder visualizar os KPIs da minha cantina
P1
Indicadores Chave de Desempenho (KPIs)
ENGREQ_51 #19
Como gestor da cantina quero poder visualizar os KPIs da minha cantina
P1
Indicadores Chave de Desempenho (KPIs)
Add item
In progress 1 Estimate: 0
This is actively being worked on
ENGREQ_51 #21
Como administrador de rede de biocantinas quero visualizar os KPIs de toda a rede e poder refinar por cantina, refeitório e produtor
P1
Indicadores Chave de Desempenho (KPIs)
Add item
In review 1 Estimate: 0
This item is in review
ENGREQ_51 #25
Adicionar estado "Aguardar plano de produção" aos menus
P1
Planeamento de Refeições
Add item
Done 17 Estimate: 20
This has been completed
ENGREQ_51 #22
Como gestor de cantina/administrador de rede de biocantinas quero colocar um fornecedor em quarentena, de forma que os produtos deste fornecedor nao entrem na elaboracao das refeicoes enquanto a quarentena estiver ativa.
P1
2
S
Quarentena
ENGREQ_51 #24
Supplier application form - Validar localização do produtor via seleção de freguesia
P1
M
Contratação de Forncedores
ENGREQ_51 #23
Como gestor de cantina/administrador de rede de biocantinas quero ativar fecho sanitário numa freguesia (bloquear produtos dos fornecedores)
P1
Quarentena
ENGREQ_51 #7
Como nutricionista quero poder alterar e aprovar a informação nutricional e ingredientes alergénios de uma receita
Add item
Block 0 Estimate: 0
Add item

The Product Backlog continued to be managed through a Kanban board with five workflow states: Backlog, In progress, In review, Done, and Block. This structure allows the team to clearly track each item's progress from planning to completion.

Organization by Epics:

Backlog items were categorized using Epics, which represent the main functional areas of the system. Based on the Sprint 2 board, the epics used include:

- Quarantine – sanitary control actions such as placing a supplier in quarantine or applying sanitary shutdown rules that prevent products from being used in meal preparation.
- Key Performance Indicators (KPIs) – KPI access and visualization at different levels (e.g., canteen and cafeteria, and network-wide refinement).
- Meal Planning – workflow improvements related to menu management (e.g., adding the status "Waiting for production plan").
- Supplier Contracting – improvements to the supplier application process, such as validated producer location selection (e.g., freguesia-based registration).

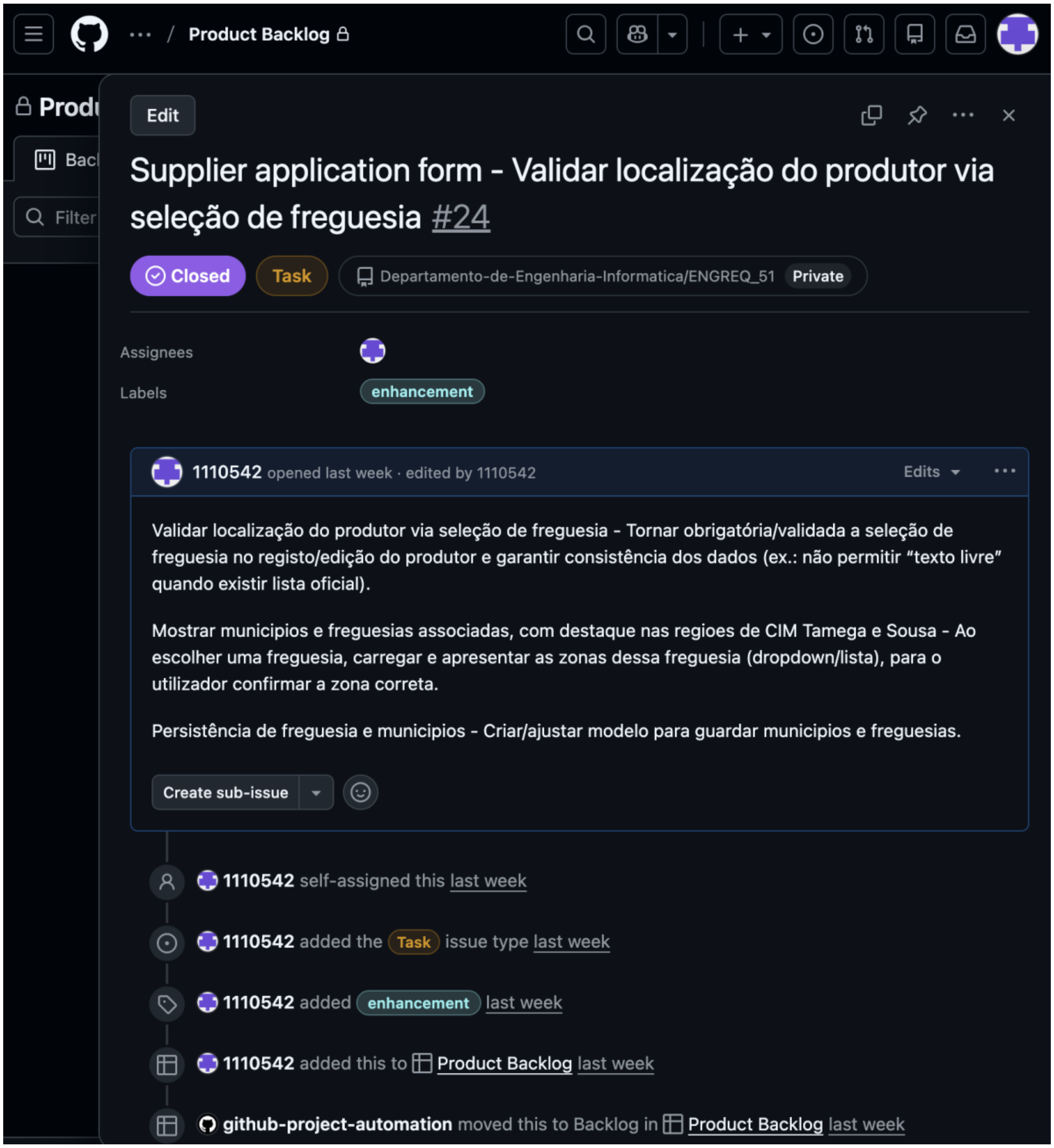
This epic-based organization helps maintain functional coherence across features, clearly visualize progress by functional area, and support informed strategic and prioritization decisions.

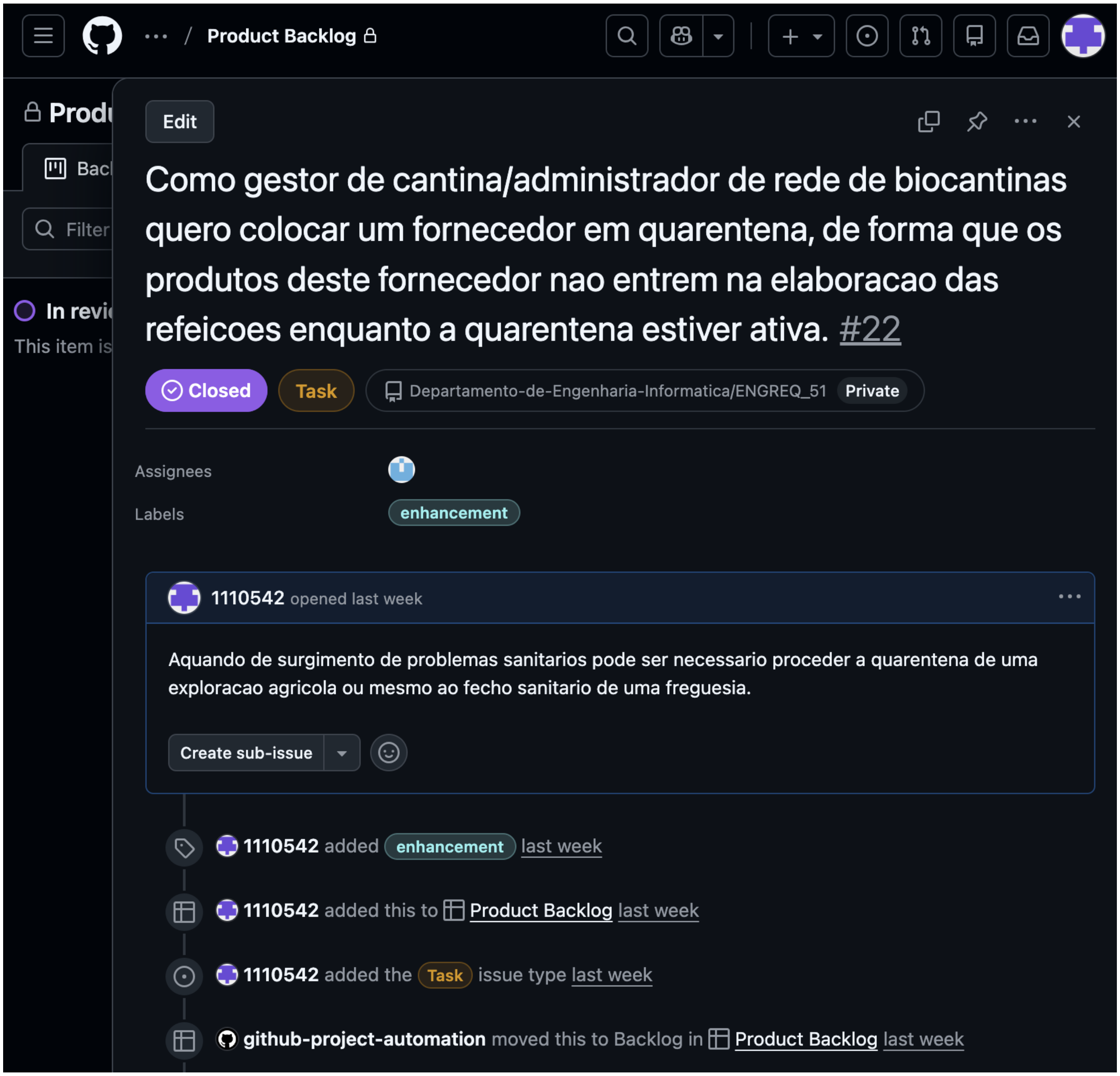
Priority and effort estimation

In addition to being organized by epics, the product backlog also includes effort size estimates and priority levels for each item. The size estimation is represented by XS (Extra Small), S (Small), and M (Medium), allowing the team to assess complexity and effort more accurately for sprint planning and capacity management.

Furthermore, backlog items are assigned priority levels such as P1, P2, and others, ensuring that the most critical and high-impact features are addressed first. This combined approach to prioritization supports effective planning, improves decision-making, and helps align development efforts with business and project goals.

In Sprint 2, we also used GitHub labels to track feedback related work. Tasks created to address comments from the Sprint 1 presentation were tagged with **"enhancement"**, so they can be easily identified and followed on the board.


Product Backlog
Prod
Bac
Filter
Edit
Supplier application form - Validar localização do produtor via seleção de freguesia #24
Closed
Task
Departamento-de-Engenharia-Informatica/ENGREQ_51
Private
Assignees
Labels
enhancement
1110542 opened last week · edited by 1110542
Edits
Validar localização do produtor via seleção de freguesia - Tornar obrigatória/validada a seleção de freguesia no registo/edição do produtor e garantir consistência dos dados (ex.: não permitir "texto livre" quando existir lista oficial).
Mostrar municipios e freguesias associadas, com destaque nas regioes de CIM Tamega e Sousa - Ao escolher uma freguesia, carregar e apresentar as zonas dessa freguesia (dropdown/lista), para o utilizador confirmar a zona correta.
Persistência de freguesia e municipios - Criar/ajustar modelo para guardar municipios e freguesias.
Create sub-issue
1110542 self-assigned this last week
1110542 added the Task issue type last week
1110542 added enhancement last week
1110542 added this to Product Backlog last week
github-project-automation moved this to Backlog in Product Backlog last week


Product Backlog
Edit
Como gestor de cantina/administrador de rede de biocantinas quero colocar um fornecedor em quarentena, de forma que os produtos deste fornecedor nao entrem na elaboracao das refeicoes enquanto a quarentena estiver ativa. #22
Closed
Task
Departamento-de-Engenharia-Informatica/ENGREQ_51
Private
Assignees
Labels
enhancement
1110542 opened last week
Aquando de surgimento de problemas sanitarios pode ser necessario proceder a quarentena de uma exploracao agricola ou mesmo ao fecho sanitario de uma freguesia.
Create sub-issue
1110542 added enhancement last week
1110542 added this to Product Backlog last week
1110542 added the Task issue type last week
github-project-automation moved this to Backlog in Product Backlog last week

As we can see in these two backlog user stories, they are well documented and follow good project management and software engineering practices. Each requirement is clearly defined with a concise and meaningful title, making its purpose immediately understandable.

Both user stories include clear and explanatory descriptions. This level of detail helps ensure a shared understanding between stakeholders and the development team, reducing ambiguity and supporting efficient development.

7 Non-Functional Requirements

The non-functional (quality) requirements considered in Sprint 1 are those identified in the previously submitted SRS, complemented by feedback received during presentations and interactions with relevant stakeholders. In addition, BioCantinas project partners imposed two key constraints: the use of free/open-access technologies and adherence to REST design principles. The following requirements were addressed in Sprint 1, with concrete examples from what was implemented.

- NFR1: Free/Open-Access Technologies
    - Requirement: Use technologies, languages, and platforms of free access (no paid licenses).
    - Implemented during Sprint 1: The solution was built with open technologies such as React + Vite (frontend), Node.js + Express (backend), MySQL (database), and GitHub for version control and collaboration.

- NFR2: REST-Based Architecture and Interoperability
    - Requirement: The solution must follow REST design principles.
    - Implemented during Sprint 1: Functionality is exposed through REST APIs consumed by the frontend via HTTP (e.g., login/register endpoints, menus retrieval, orders and reservations operations), with standard request/response patterns and separation between UI and backend services.

- NFR3: Usability and Role-Oriented UX
    - Requirement: The UI should be understandable and support user workflows effectively.
    - Implemented during Sprint 1: Dedicated screens were created for different actors, such as students (view menus and place reservations), nursing homes (view nutritional information), canteen staff (validate reservations), suppliers (view/manage supplier orders), stock manager (manage orders), and network

manager (rank suppliers by priority). The navigation and pages were iteratively improved with UI adjustments (e.g., layout and Bootstrap fixes).

- NFR4: Accessibility Baseline
  - Requirement: Provide a baseline of accessible and clear information presentation, enabling future improvements. Implemented during Sprint 1: Menus, allergens, and nutritional values were displayed using consistent page structures and readable components (e.g., nutritional information views for nursing homes and menu detail pages), ensuring critical information is visible and understandable.

- NFR5: Performance and Responsiveness
  - Requirement: The application should remain responsive during normal operation.
  - Implemented during Sprint 1: The frontend was implemented as a single-page application (React) with client-side routing (React Router) to avoid full reloads between screens, and API calls were handled through Axios to load only the necessary data per view (e.g., weekly menu navigation, orders tabs).

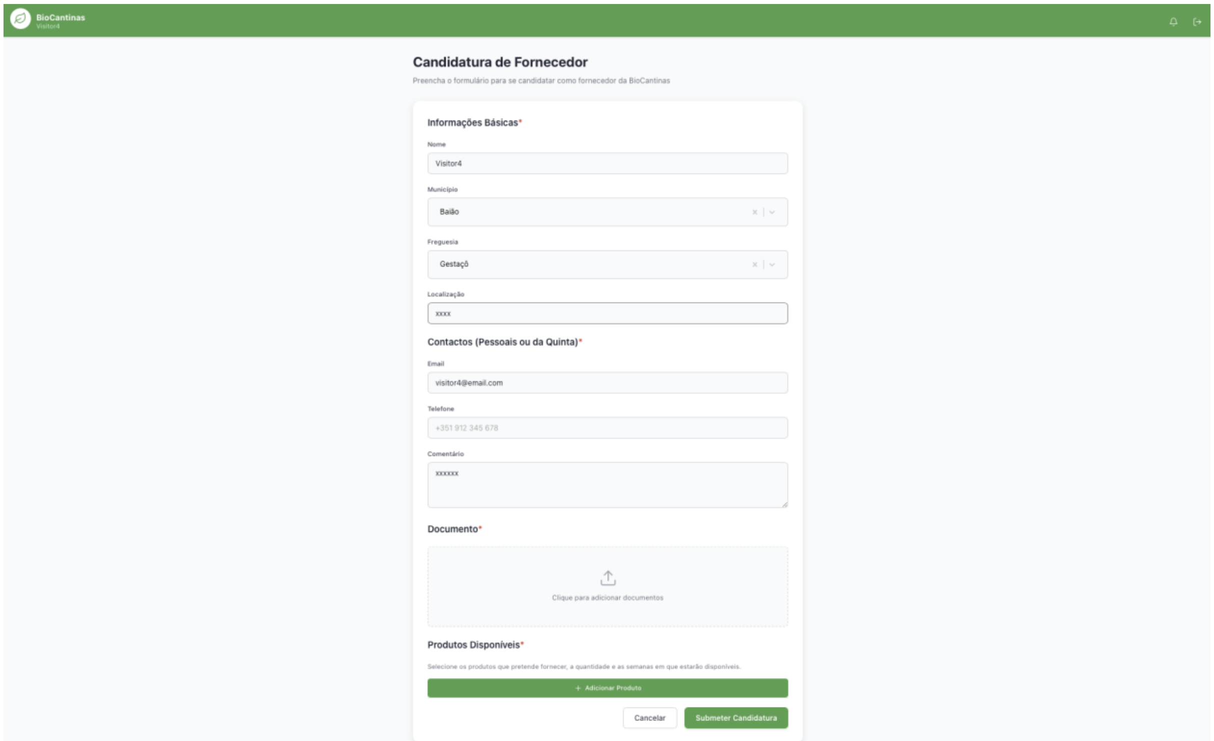
- NFR6: Security and Access Control
  - Requirement: Only authorized users should access protected features.
  - Implemented during Sprint 1: Authentication flows were implemented (e.g., login/register endpoints) and role-based behaviour was introduced (e.g., different dashboards and screens depending on user type, such as network manager vs supplier vs student).

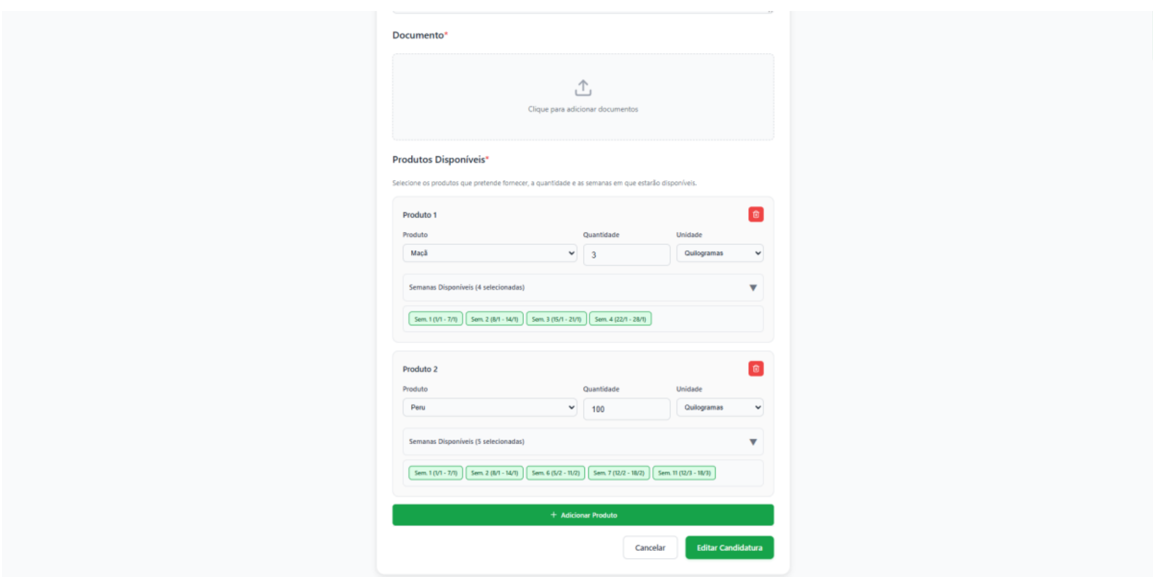
- NFR7: Reliability and Data Integrity
  - Requirement: Ensure consistent data persistence and relationships between entities.
  - Implemented during Sprint 1: The system persisted core entities in a relational database (MySQL), supporting consistent links across flows such as "applications" to "accepted applications" to "suppliers", and the ordering/provisioning features relying on products, quantities, and units.

- NFR8: Maintainability and Evolvability
  - Requirement: The solution should be easy to extend in Sprint 2 and beyond.
  - Implemented during Sprint 1: The project follows a separation of concerns: frontend (React), backend (Express REST APIs), and database (MySQL/Sequelize). Development was iterative with frequent commits/merges, and refactors were performed to improve structure (e.g., adjustments to screens, unit conversion handling, and error fixes).

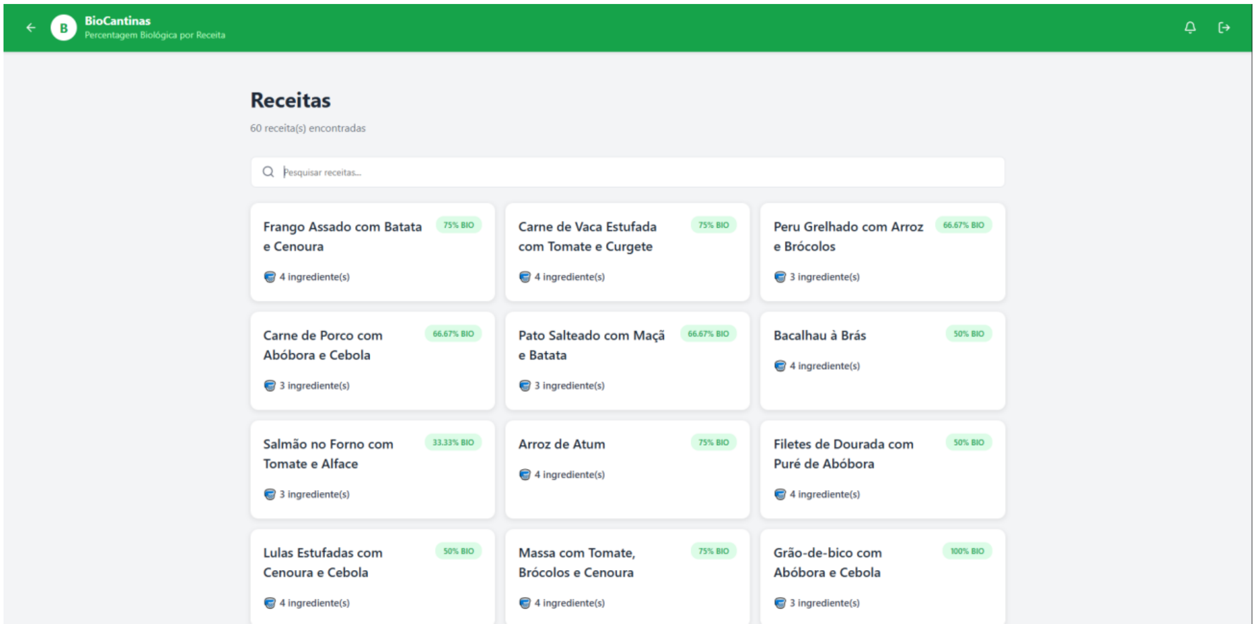
- NFR9: Portability and Reproducibility
  - Requirement: Team members should be able to set up and run the project consistently.
  - Implemented during Sprint 1: The repository includes an updated README with setup instructions, login credentials, and available routes, enabling repeatable local deployment for development and testing.

8 Application Visual Evidences

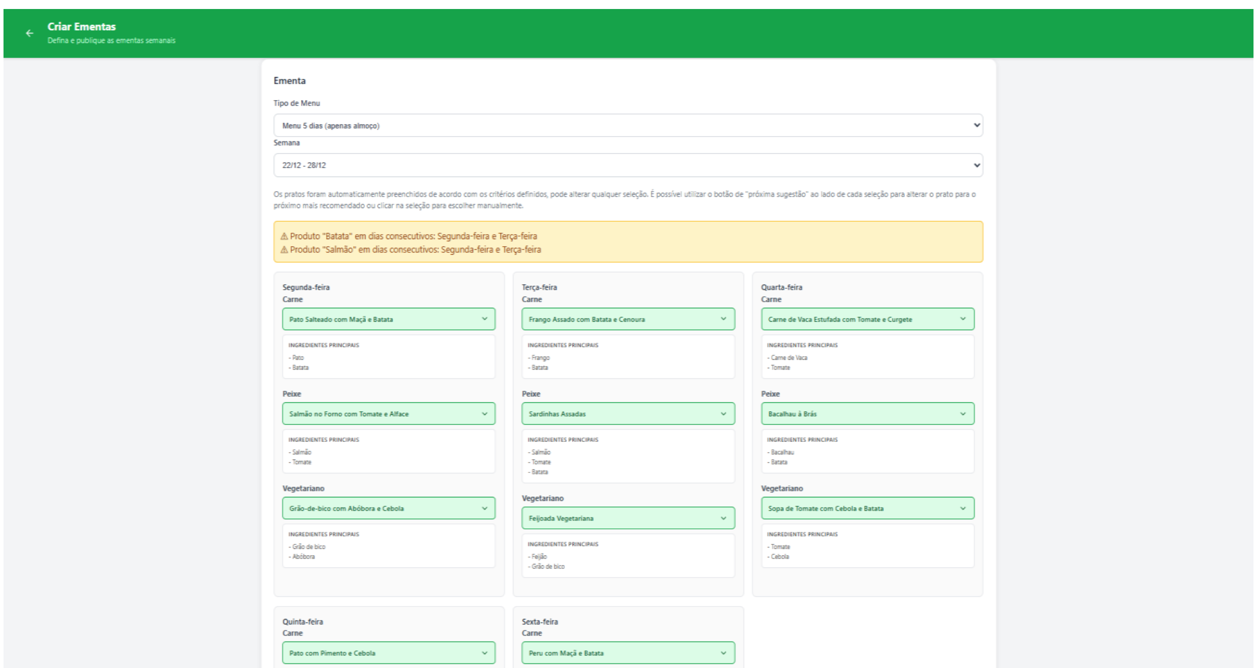


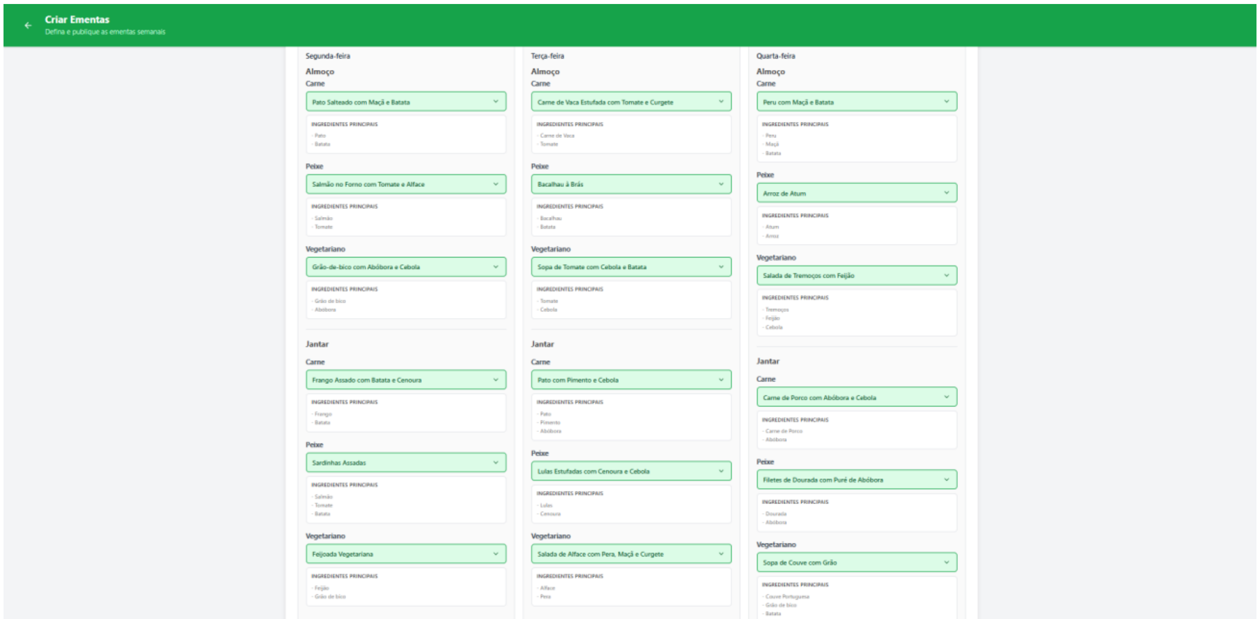


Screen Page for the application of the Supplier containing all the key information for the contract plus his detailed product capacity.

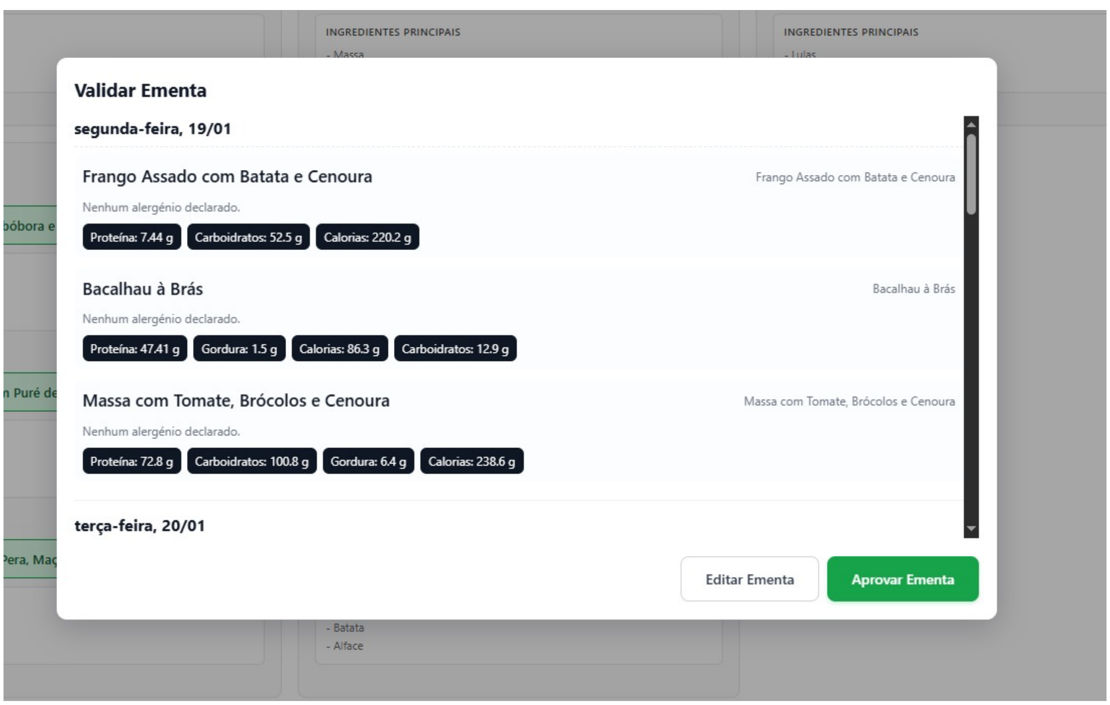


Screen Page for Recipes available and containing all details.

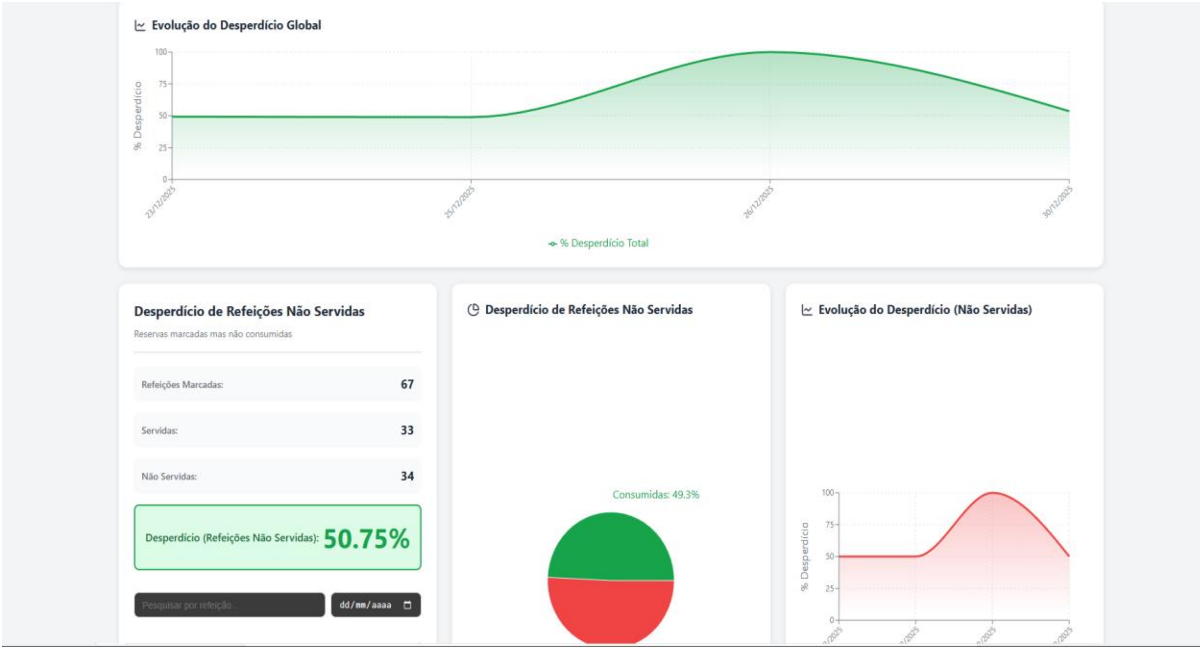


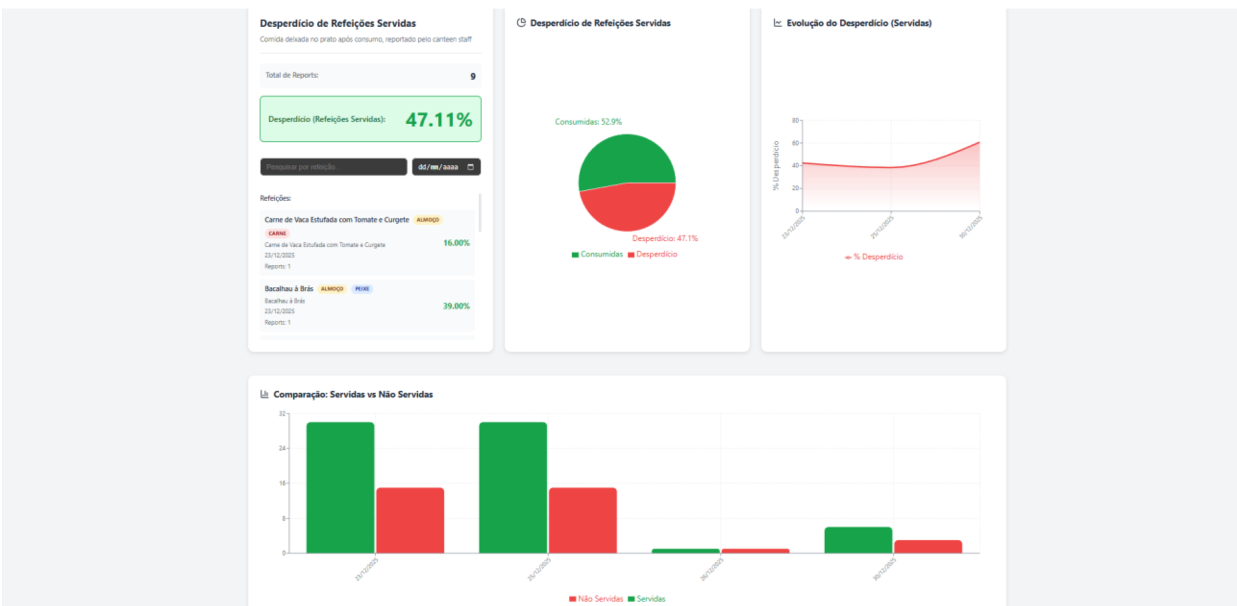


Screen Page for all Ementa creation, allowing us to choose either a 7 day week (with dinner) or a 5 day week (for schools). The week for the Ementa can be selected. The meals present are automatically generated based on a suggestive algorithms. There are warnings to prevent using the same main product multiple times in the same menu.

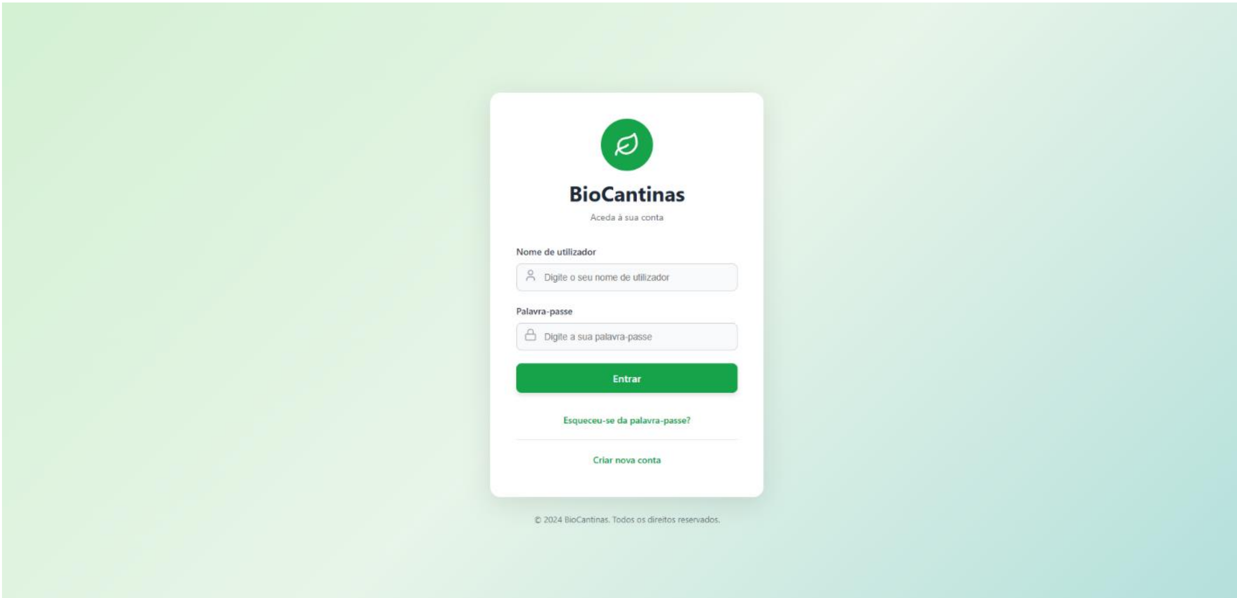


Screen Page for all Ementa Edition/ Approval containing nutritional info.

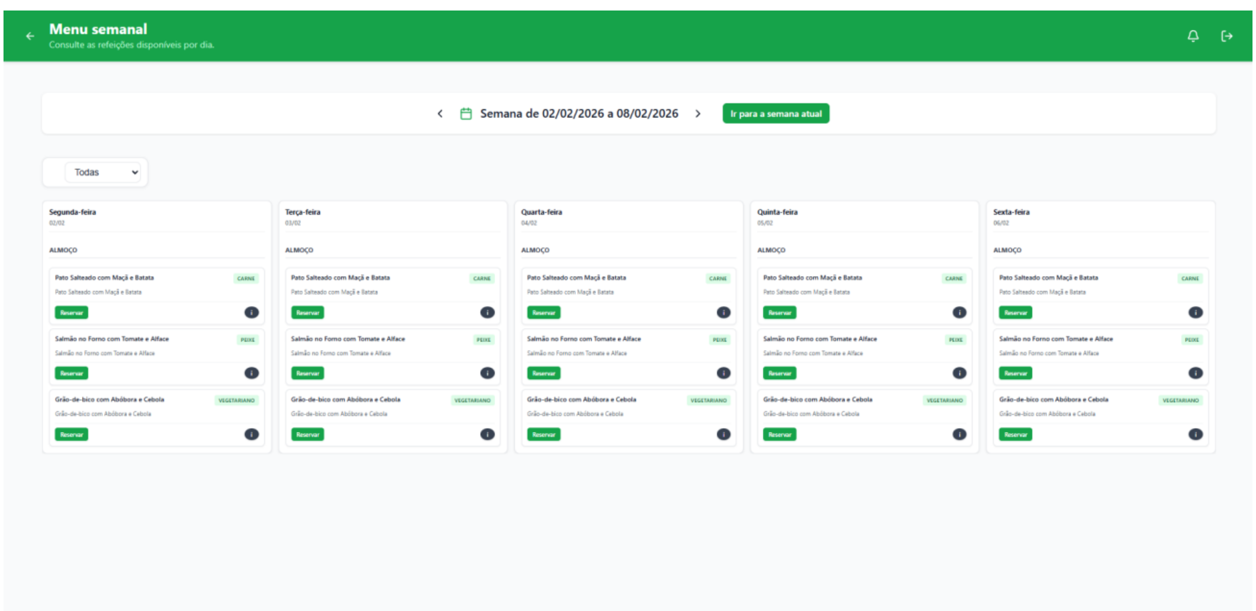




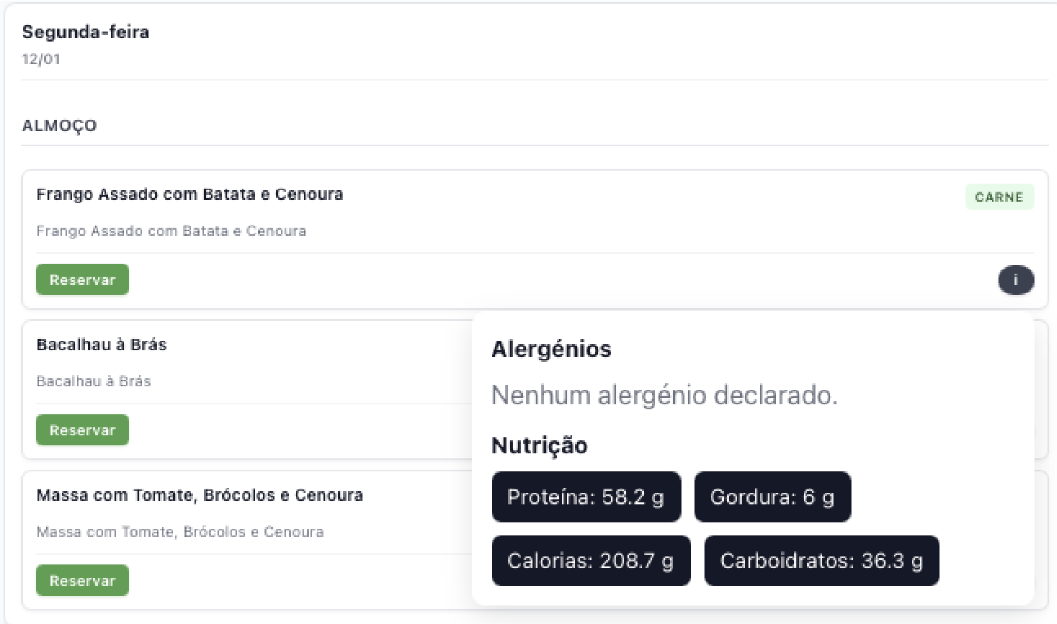
Screen Page for KPIs and Waste information with the possibility to filter by meal type, dish type and time interval.



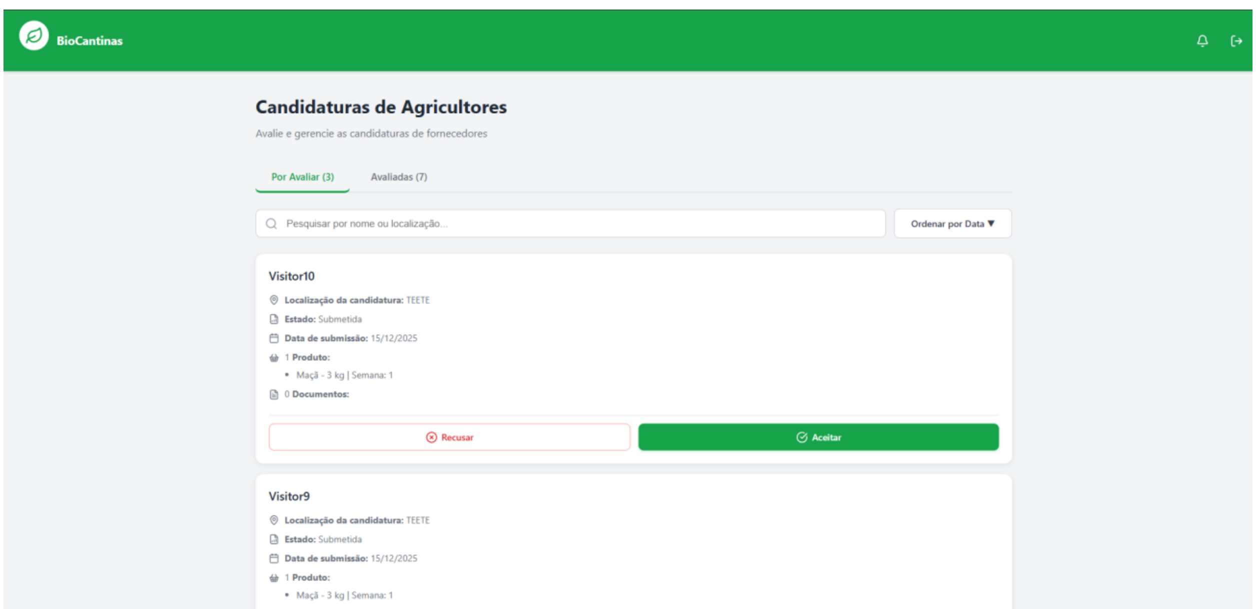
Screen Page for login in BioCantinas.



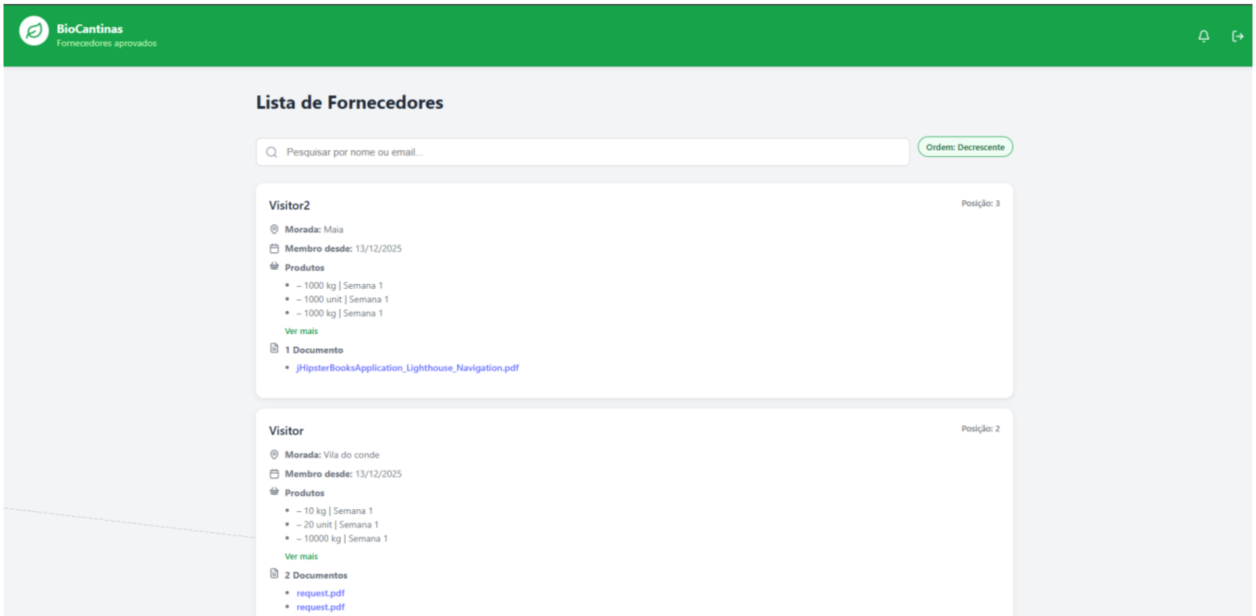
Screen Page for Menu and Respective reservation (possibility to choose the week or by default the current one) of meals.



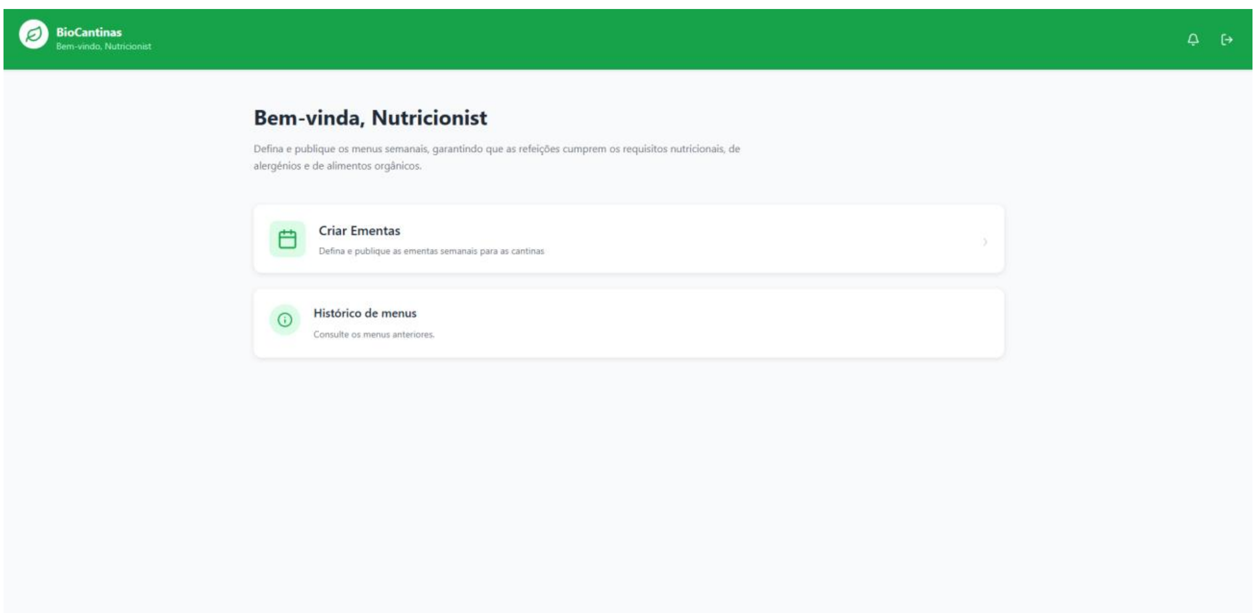
Screen Page for Menu Focusing on the nutritional information of a Meal.



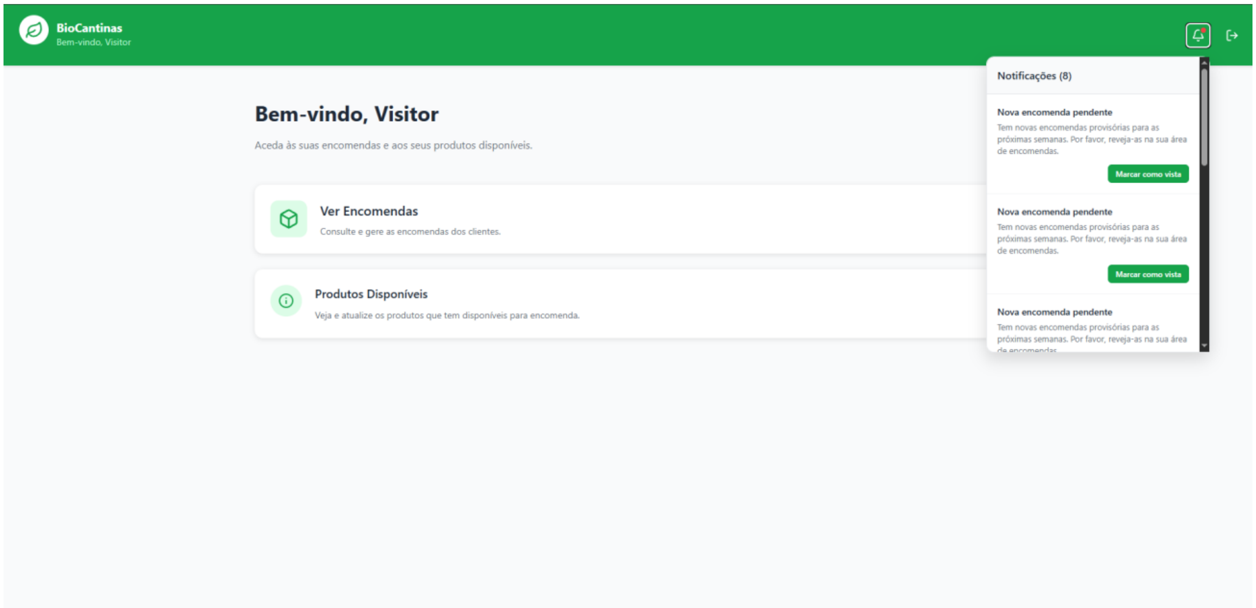
Screen Page for Applications evaluation of the farmers.



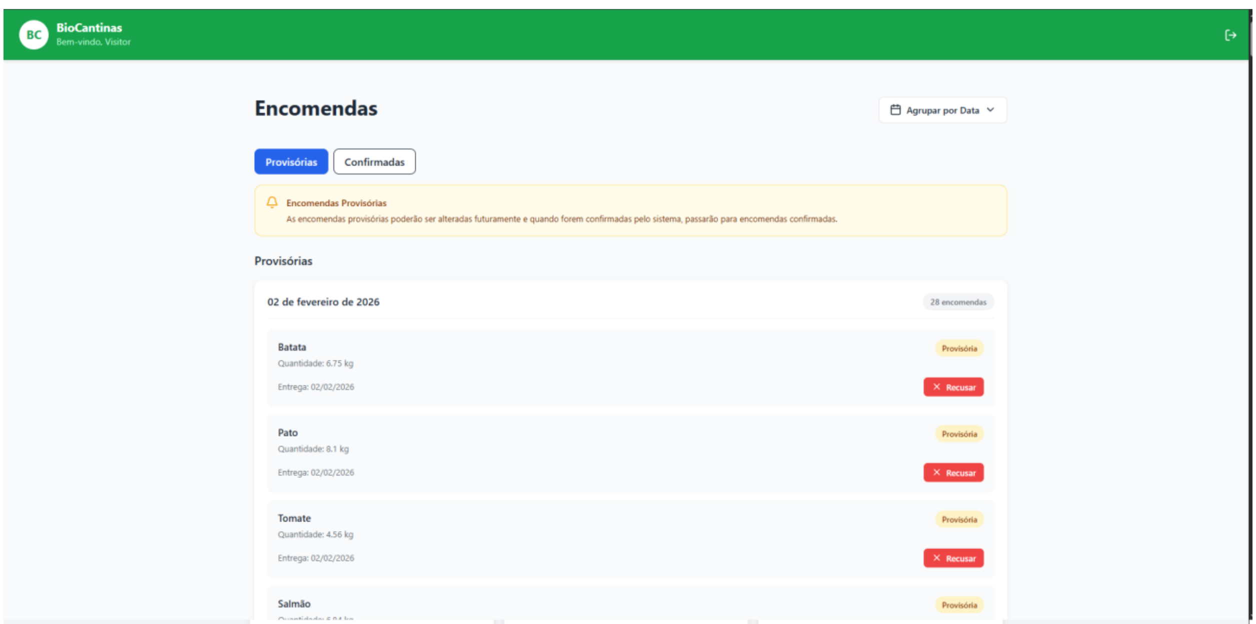
List of the system suppliers ordered by application date with a filter of name and e-mail.



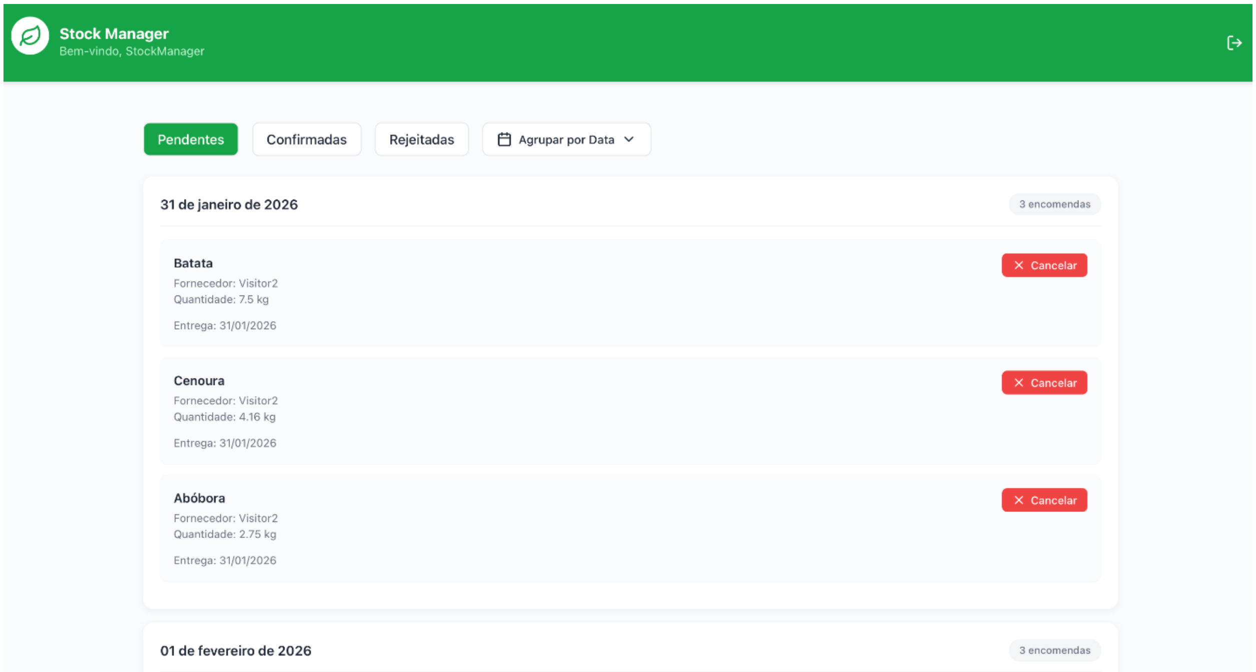
Screen Page for home of a user with the role Nutricionist, it contains the Ementa creation and the Menu history



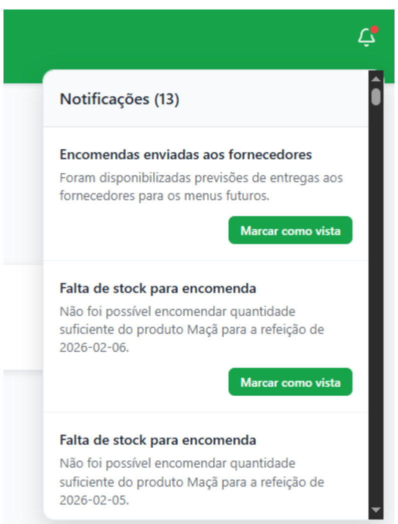
Screen Page for home of a user with Role Supplier, it contains the notifications for the Orders, the check of notifications and the available products.



Screen page for existing Orders of the supplier (Visitor PoV).

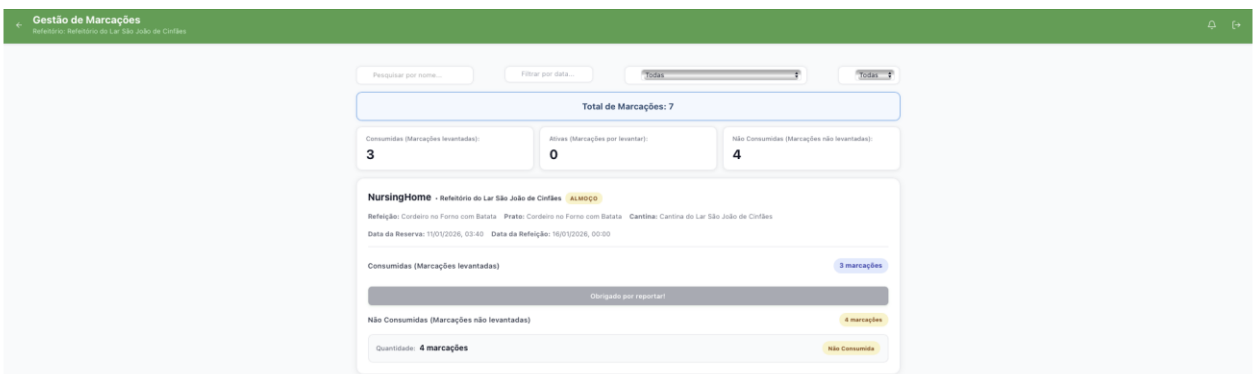


Screen page for existing Orders (Stock Manager PoV) with filter by Data or Product

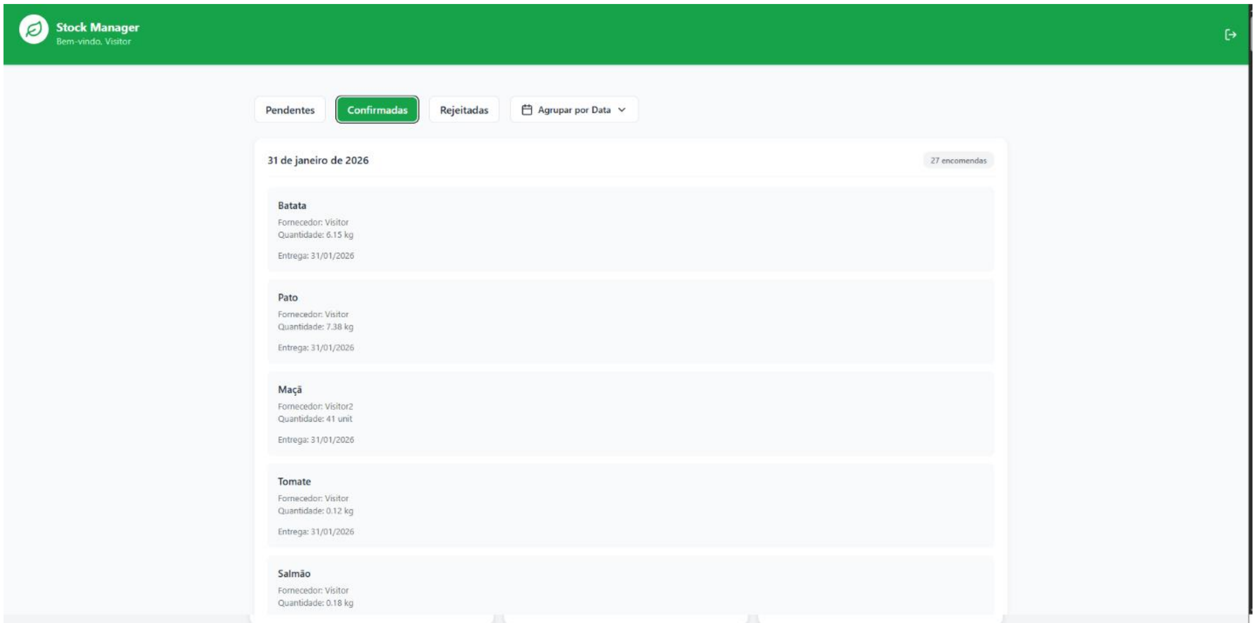


Partial Screen with notification for a stock manager about orders.

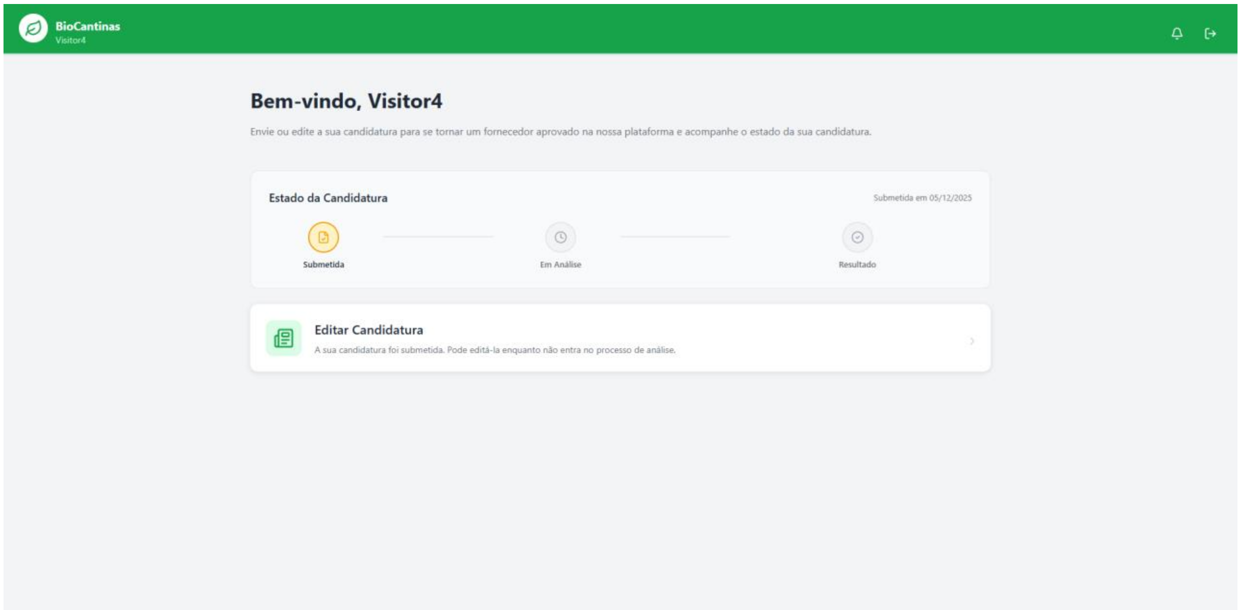
Partial Screen with confirmation when reservation of a meal.

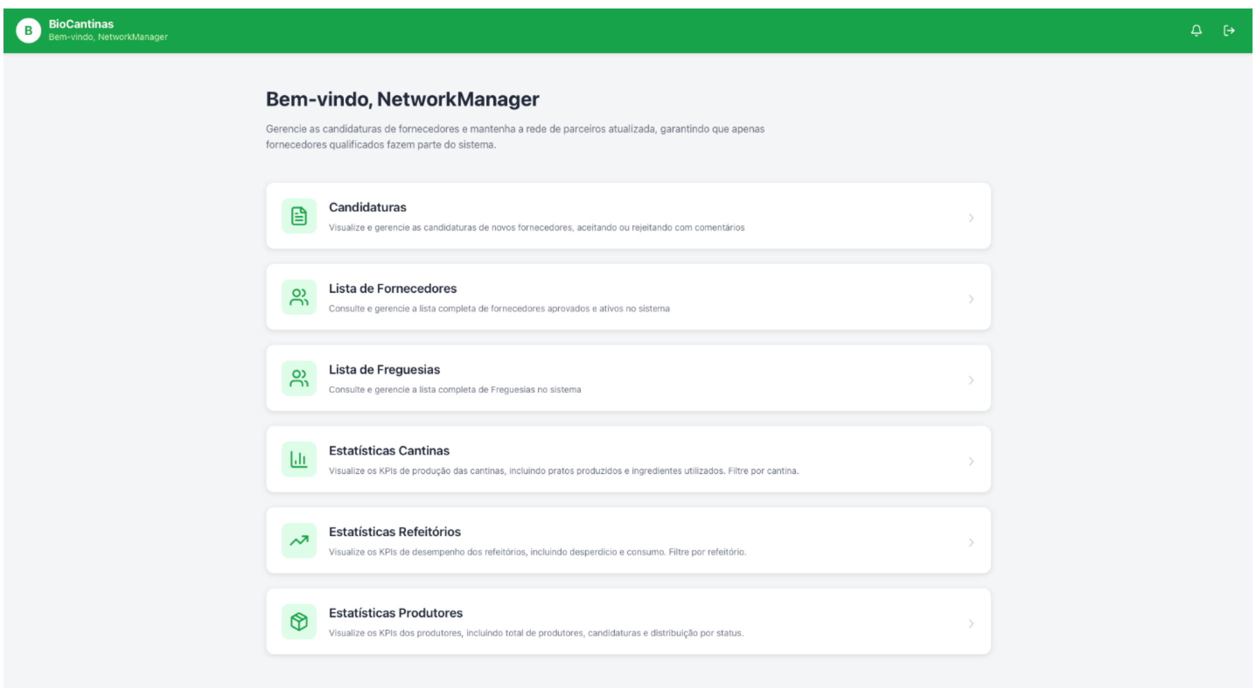


Screen for Management of Bookings.

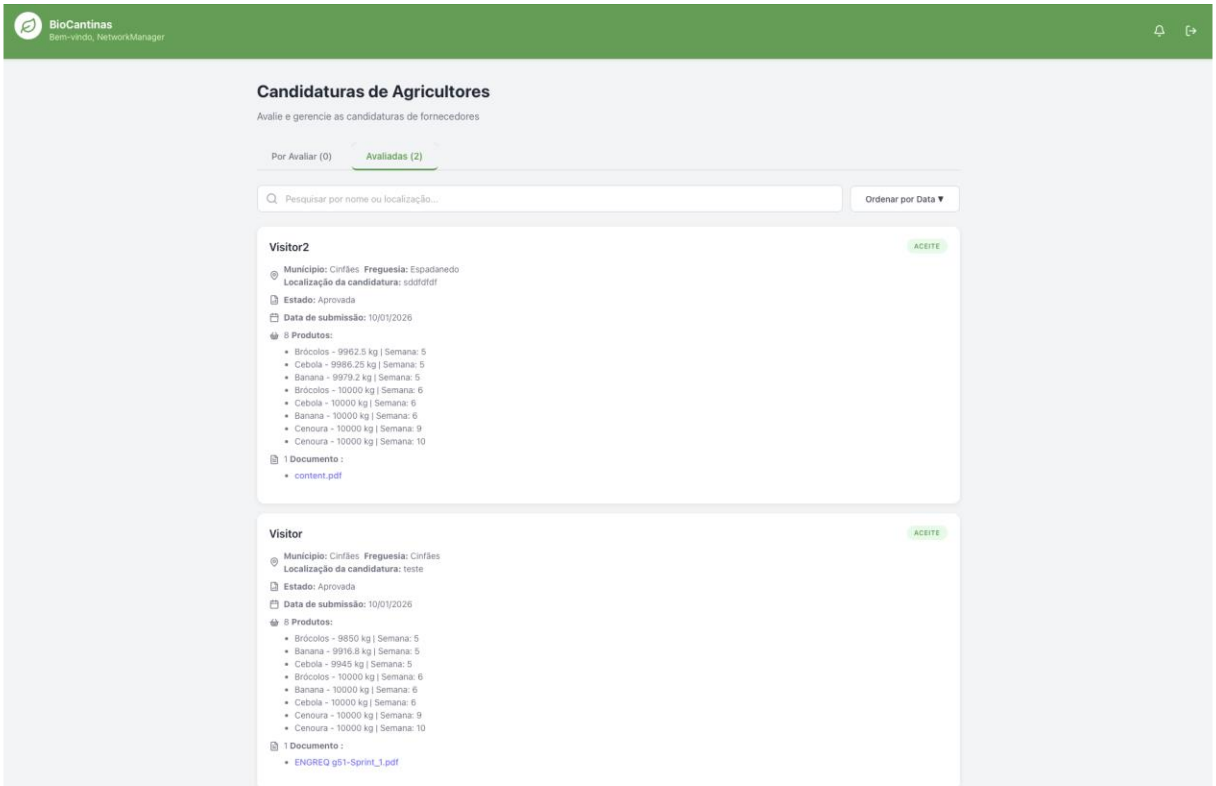


Screen for Stock Manager.

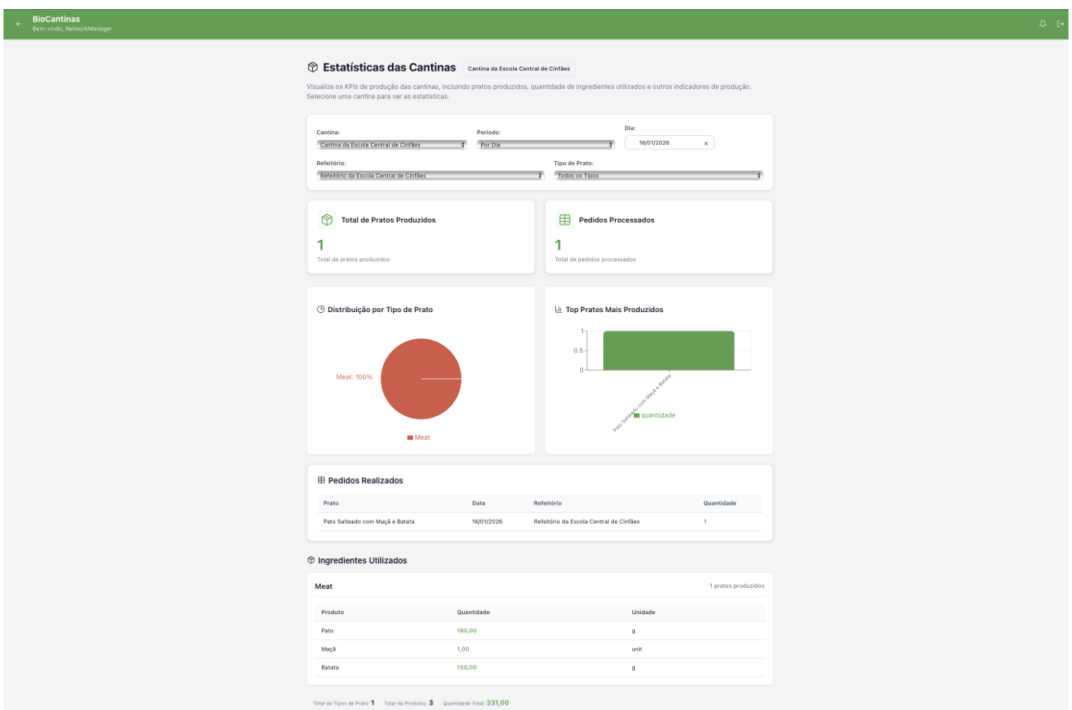




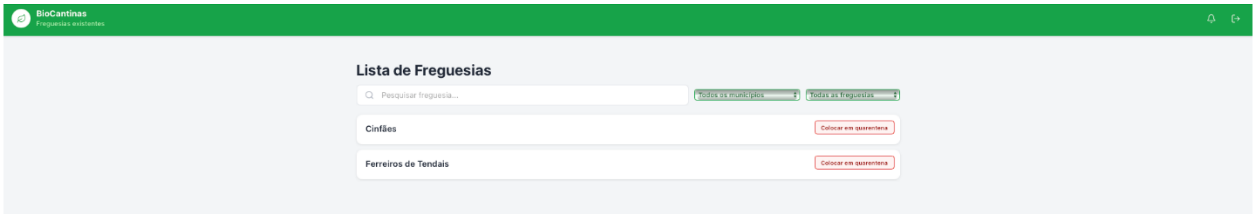
Screen Page for home of a user with the role Network Manager, it contains the Candidaturas, the Lista de Fornecedores, the Lista de Freguesias. The Estatisticas Cantinas, the Estatisticas Refeitório and the Estatististicas Produtores.



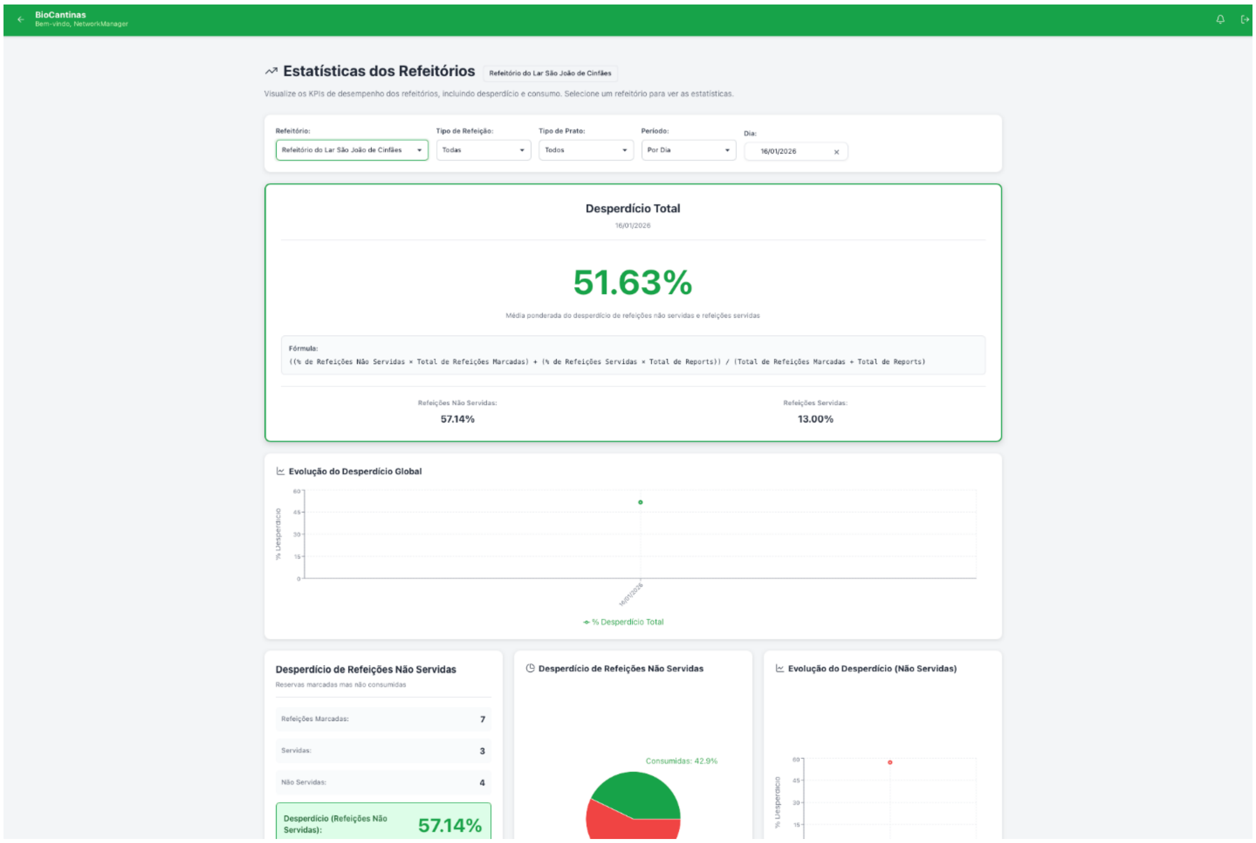
Screen Page for Applications (Network Manager PoV)



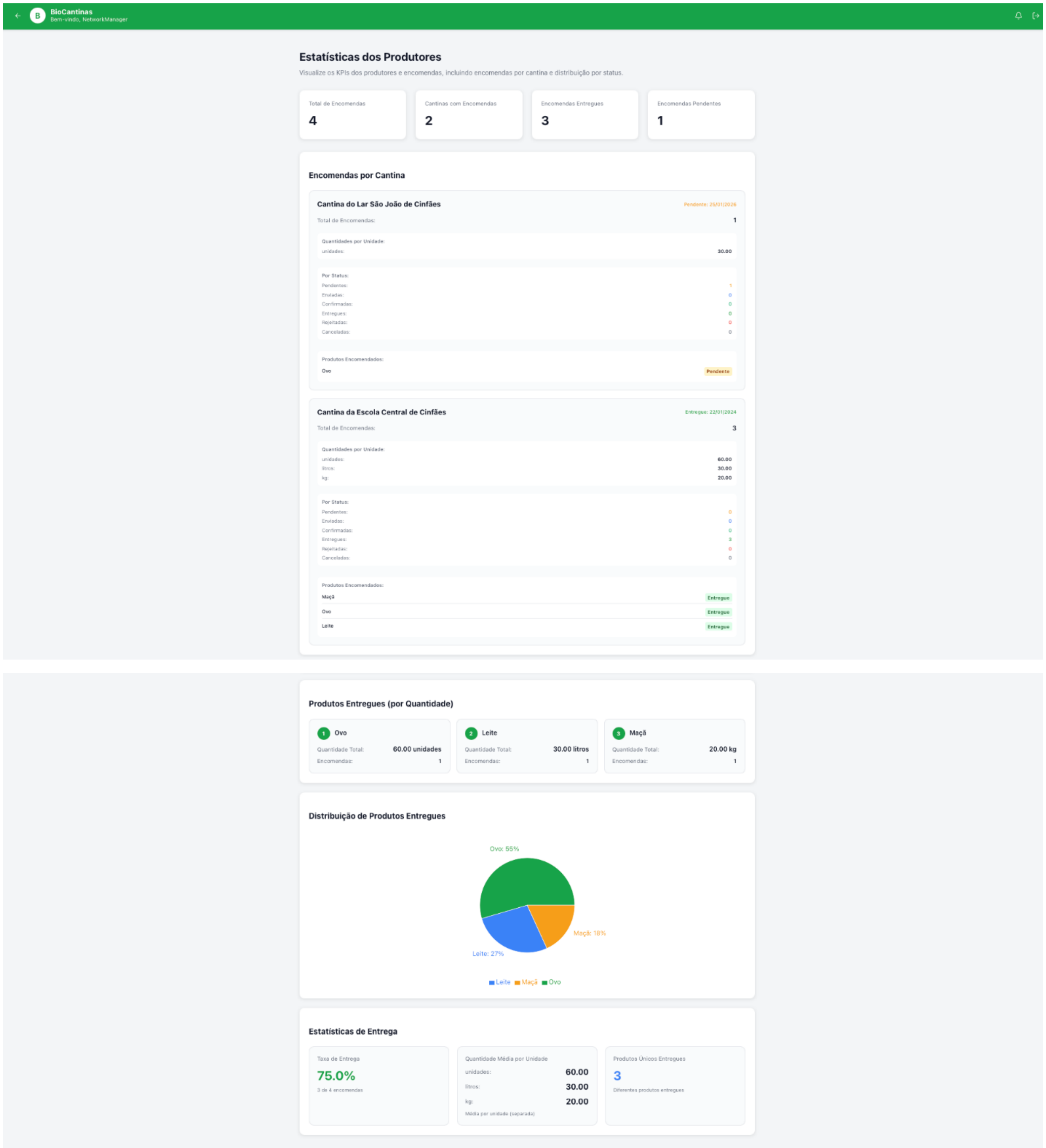
Screen for Canteen Stats (Network Manager PoV)



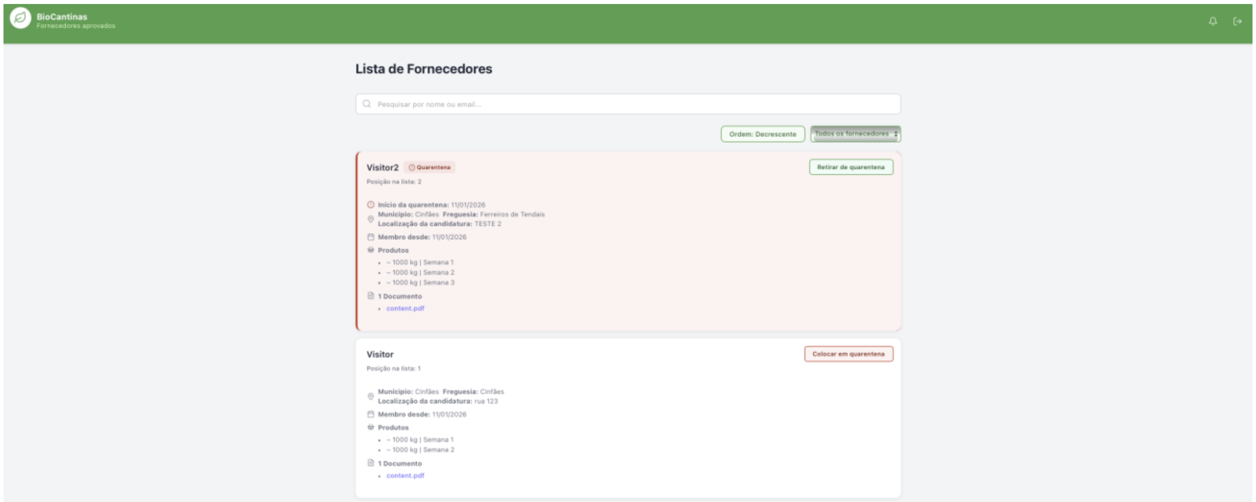
Screen Page for Freguesias List (Network Manager PoV)



Screen Page for Refeitório Stats (Network Manager PoV)



Screen Page for Producer Stats (Network Manager PoV)

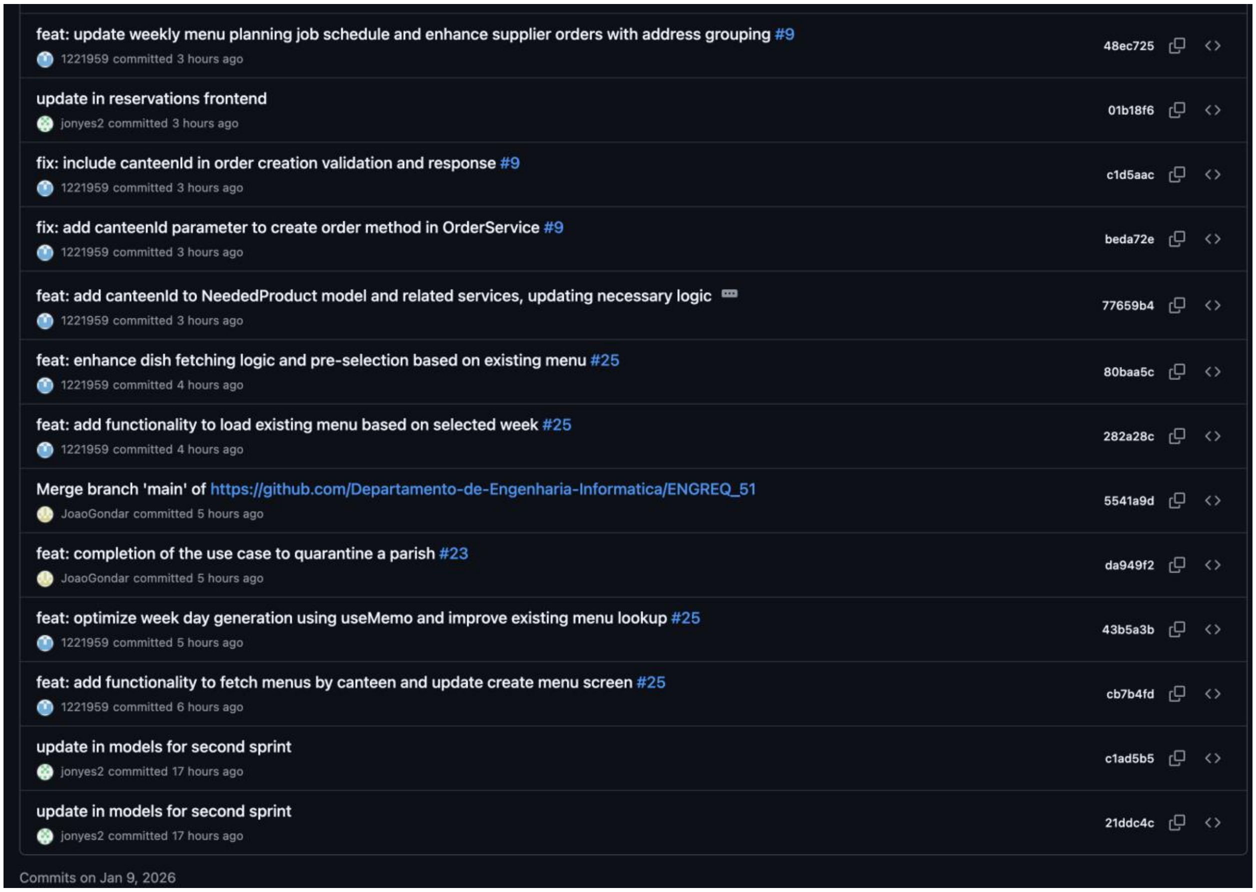


Screen Page for Suppliers list (one supplier in quarenteen).

9 Version Control Evidence

The version control evidence shows how the team used the repository to manage changes, track progress over time and ensure that implemented features and fixes were properly recorded and could be reproduced. Git and GitHub were used to collaboratively manage the codebase throughout the sprint. The team frequently committed changes to ensure traceability of progress, integrated work through merges and used the repository as the single source of truth for the application and supporting documentation (e.g. the README file and setup instructions).

The following figure presents screenshots from the project repository showing the commit history and activity during Sprint 2, and demonstrating the incremental implementation of features, bug fixing and refactoring.



The complete list of commits can be found at https://github.com/Departamento-de-Engenharia-Informatica/ENGREQ_51/commits/main

10 Conclusion

Overall, the requirements defined for the project were implemented and validated in line with stakeholder expectations, resulting in a complete and functional application by the end of Sprint 2. Although the solution fulfils the MVP goals, there is still scope for future improvement and expansion to address additional stakeholder needs and emerging requirements. In addition, several complementary features were implemented beyond the initial specification to strengthen the functionality and improve the usability of the final product.